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How informative are insider trades and analyst recommendations?[☆]

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A B S T R A C T

This study evaluates the interactions between, and informativeness of, insider trading and analyst recommendations. We find that analyst recommendations significantly affect subsequent insider trading, but not vice versa. Surprisingly, in aggregate, insiders buy more shares following analyst downgrades and sell more shares following upgrades. This pattern persists even after controlling for analysts' momentum and insiders' contrarian trading preferences. Analysts, in contrast, do not systematically take into account insider trading when revising their recommendations. More importantly, we show that these two information signals complement each other although insider buying could be a singularly strong signal that substitutes the informativeness of analyst recommendations under certain circumstances. Overall, our findings highlight the important dynamics and financial market consequences between the two crucial groups of information providers.

1. Introduction

Insider trades and analyst stock recommendations are perhaps the two most salient information signals to investors. A vast literature has provided robust evidence that insiders exploit their informational advantage to earn abnormal returns on their trades (e.g., Jaffe, 1974; Finnerty, 1976; Seyhun, 1986; Damodaran and Liu, 1993; Rozeff and Zaman, 1998; Lakonishok and Lee, 2001; Jeng et al., 2003; Brochet, 2010). Similarly, another strand shows that financial analysts are generally knowledgeable about the firms and industries they follow, and their recommendations could generate investment value.[1] While insider trading and analyst recommendations have been studied extensively in the literature, two critical, yet less explored, issues are the extent to which the informativeness of one informed group's activity is affected by the other group's, and more importantly, their relative strengths in providing value-relevant information to other market participants. This research fills the void by studying these two issues. Examining the latter is crucial as it advances our understanding of the incremental role of their actions in disseminating information in the financial market.

Theoretically, the information contents of insider trades and analyst recommendations could substitute for each other. Fishman and Hagerty (1992) and Khanna et al. (1994) postulate that active insider trading could adversely affect informed outsiders such as analysts from acquiring valuable information. As a result, analysts' incentives could be 'crowded out', and their recommendations become less informative. Interestingly, recent research suggests that the crowd-out effect could also go the other way. That is, if analysts act as effective information intermediaries, their activities could reduce insiders' trading incentives and render their trades less informative. For instance, insiders could receive feedback from their firms' share prices in the secondary market to adjust their trading (Khanna et al., 1994). In addition, a firm with more analyst following is associated with lower insider buying and smaller profitability of insider trades (Frankel and Li, 2004), while the loss of analyst coverage is associated with more profitable insider trades (Kelly and Ljungqvist, 2008; Wu, 2016; Ellul and Panayides, 2018). Overall, this substitution view posits that

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[1] As in most studies, we focus on sell-side, rather than buy-side, analysts. Stickel (1995); Womack (1996); Barber et al. (2001); Jegadeesh et al. (2004), and Loh and Stulz (2011) find that analyst recommendations, especially downgrades or recommendations from leader or star analysts, are informative. Altinkilic and Hansen (2009), on the other hand, argue that analysts piggyback on recent corporate news and events in their recommendations. Bradley et al. (2014); Li et al. (2015), and Yezegel (2015), however, show that recommendation revisions facilitate price discovery regardless of confounding corporate news.

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when one group plays a dominant role in disseminating information, the other group's role becomes inconsequential regardless of whether it concurs or disagrees with the assessment of the dominating group. As a result, the stock market reactions mainly reflect the dominating signal.

In this study, we propose an alternative view that the information contents of insider trading and analyst recommendations complement each other. The reason is threefold. First, one group's activity could motivate, rather than disincentivize, the other to disclose its own information or trade on the information it receives. For instance, insider trading, particularly buying, is associated with lower analysts' earnings forecast errors (Lustgarten and Mande, 1995; 1998; Sivakumar and Vijayakumar, 2001). Loureiro and Taboada (2015) show that the improved information environment post-IFRS enhances insiders' ability to learn from analysts. Alldredge and Cicero (2015) further document that insider selling, motivated by public information (e.g., analyst recommendations), could be informative. Second, insiders and analysts could reveal different aspects of information. Specifically, prior research indicates that insiders have ready access to firm-specific private information, whereas analysts also gather and process industry and market information (e.g., Piotroski and Roulstone, 2004; Howe et al., 2009; Hutton et al., 2012; Kadan et al., 2012; Wang, 2019). Third, insiders and analysts face different constraints in revealing their information to the market. In particular, although insiders are considered most informed among informed investors, insider trading regulations and associated litigation risk can deter them from signaling their private information through their trades, especially in periods of high information uncertainty (Huddart et al., 2007; Marin and Olivier, 2008). Analysts, in contrast, tend to increase their information production and become more desirable in those high uncertain periods (Yezegel, 2015; Loh and Stulz, 2018). Besides, insider buying tends to be more informative than selling as the latter could be due to liquidity shocks or diversification demands (e.g., Lakonishok and Lee, 2001; Kallunki et al., 2009; Brochet, 2010). In contrast, analyst downgrades, in general, are more credible than upgrades due to the findings that their optimistic recommendations are associated with conflicts of interest, and they are reluctant to issue downgrade recommendations unless necessary so as to foster a good relationship with the management of firms they cover and access private information (e.g., Francis and Philbrick, 1993; Das et al., 1998; Lin and McNichols, 1998; Michaely and Womack, 1999, Lim, 2001; Cowen et al., 2006; Ljungqvist et al., 2007; Kolasinski and Kothari, 2008). Together, our proposed complementarity hypothesis asserts that the information contents of insider trading and analyst recommendations may be incomplete and could provide complementary signals to the market. This proposition implies that when the two signals concur, their joint market reaction should be stronger than each individual signal alone as one could further strengthen the other. However, when they disagree, their joint market reaction should be stronger (weaker) than the negative (positive) signal as one could attenuate the other.

We start by examining the intertemporal interactions between insiders and analysts to determine whether their activities influence each other. For the effect of insider trading on analyst recommendations, it is sensible to assume that analysts assimilate and process information in insider trading when updating their recommendations since insider trades are part of publicly accessible information. If they do, they could revise their recommendations either in accordance with or in the opposite direction of insiders' signals. The latter is likely to occur, for example, when analysts believe that insider selling is for liquidity reasons and are confident that the stock is still worth investing instead. On the other hand, analysts might not materially consider insider trading information. For instance, existing studies indicate that analysts display certain information-processing biases or fail to consider important signals (see, e.g., Bradshaw, 2004; Jegadeesh et al., 2004; Engelberg et al., 2018; Guo et al., 2019); as a result, we should expect little association between insider trading and subsequent analyst recommendation revisions.

Existing literature also puts forth three possible motives on the effect of analyst recommendations on insider trading. First, the information hypothesis argues that insiders trade in the same direction as analyst recommendation revisions, and analyst downgrades should be more informative than upgrades, provided the severity of the former. Second, the signaling hypothesis, in contrast, postulates that insiders trade in the opposite direction of recommendation revisions, and this inverse relationship should be mostly driven by the effect of analyst downgrades on insider purchases if insiders are more concerned about declining share prices. Third, the liquidity hypothesis suggests that insider selling following analyst upgrades is more likely than insider buying following downgrades. In addition, insiders could strategically move stock prices before trading by exploiting certain tactics such as managing earnings or disclosing significant news, and analysts respond to those tactics. The literature suggests that analyst recommendation revisions are associated with significant short-term share price reactions. While it is unlikely that insiders plan or time their transactions conditional on market reactions to analyst recommendation changes, their trades may be triggered by such opportunities, especially for liquidity reasons.[2]

Our results show that analyst recommendation revisions significantly affect insiders' trading decisions, but surprisingly, insiders, in aggregate, trade in the opposite directions of those revisions; insider buying (selling) significantly increases following analyst downgrades (upgrades). This contrasting pattern persists even after controlling for analysts' preference for the momentum strategy and the possibility that insiders either execute an effective contrarian trading strategy or opportunistically depress (raise) share prices before buying (selling). It is also robust to additional controls such as the degree of analyst coverage, stock and analyst characteristics, and the overall information environment. Further analysis suggests that insiders' motives are likely driven by signaling, rather than information, reasons as insider buying significantly increases following analyst downgrades, especially those with more severe return outcomes. On the contrary, while insider trading exhibits a weak influence on analyst upgrades, there is no consistent pattern of analysts exploiting insider trading information when revising their recommendations. This finding, albeit surprising, corroborates existing evidence that analysts do not incorporate specific price-sensitive signals into their recommendations.

Next, we evaluate the relative informativeness of insider trading and analyst recommendations. Our initial result suggests that both insider buying and selling, while only analyst downgrades, are informative. Consequently, we exclude analyst upgrades from this part of analysis and focus on the interactions between insider trading (buying and selling) and analyst downgrades. Overall, the find-

[2] Another possibility is due to the timing of insider trades. Note that we measure insider trading activity based on their filing dates rather than their actual transaction dates since trading information is not made public until insiders file their transaction reports with the SEC. In the early part of our sample period, a considerable lag exists between the actual transaction date and the filing date because Section 16(a) of the Securities Exchange Act of 1934 only requires that insiders file within ten days at the end of the month in which the transaction occurs. Thus, insider trades could occur around the same time as analyst recommendation revisions, but they are reported on a later date. However, as discussed previously, an amendment to Section 16(a) in the Sarbanes-Oxley Act that requires insiders to file before the end of the second business day after the transaction was executed became effective on August 29, 2002. Since our second subsample restricts the sample period from 2003 and onward and the results remain significant, this scenario is unlikely.

ings are more in line with the complementarity hypothesis. Specifically, our analysis reveals that when conditioned on analyst downgrades, stocks with more insider buying than selling experience significantly higher returns, implying that insider trading is able to produce additional valuable information beyond that by analyst recommendations. Similarly, when conditioned on insider selling, stocks with analyst downgrades are associated with lower returns. Further, we find that the joint informativeness of two concurring negative signals (insider selling and analyst downgrades) is more severe than the individual informativeness of each signal alone, whereas the joint informativeness of the two conflicting signals (insider buying and analyst downgrades) falls between the individual informativeness of negative and positive signals. Hence, the combined evidence supports the notion that both insider trading and analyst downgrades contain value-relevant information; one could effectively reinforce the other if they concur, but one could also negate the other if they disagree. Importantly, although the results on the whole lend support for the complementarity effect between insider trading and analyst recommendations, our results reveal a singular impact of insider buying. Specifically, the positive return effect of insider buying is more notable following, rather than before, analyst downgrades, and especially for large impactful downgrades. This evidence implies that insider buying following downgrades is generally motivated by signaling reasons.

We conduct additional tests utilizing three subsamples to shed more light on the complementarity vs. substitution effects of insider trading and analyst recommendations. Specifically, we only include 'top' insiders in the first subsample and limit the sample period in the second to post-2002, a period marked with material regulatory changes to enhance insider-trading disclosure and limit analyst information access. The results show that the complementarity effect is reduced in these two subsamples, suggesting that for insider trading, top insiders' buying decisions are more informative than other insiders' even though they are constrained by more stringent regulations, and for analyst recommendations, the imposed restrictions from accessing private information sources post-2002 significantly affect analysts' information production. For the third subsample, we only include observations in the months prior to and following the EA month and exploit substantial changes in the information environment during the EA periods to further examine how the informativeness of insider trading and analyst recommendations differs. Specifically, the information uncertainty heightens in the month prior to an EA and subsides following an EA. The nature of information is also different around an EA; private information regarding a firm is more prevalent prior to its EA while the public information is more dominant following its EA. We find that this subsample provides robust support for the complementarity effect between insider trading and analyst downgrades, and such effect is stronger after EAs, implying that the informativeness of analyst recommendations stems from analysts' ability to skillfully analyze public information. However, the evidence also indicates more singular dominance of insider buying following analyst downgrades, and such effect is more prominent before EAs, implying that insiders exploit their private information advantage. Collectively, our findings suggest that except for the case in which strong signaling of insider buying could severely affect the informativeness of prior analyst downgrades, the information contents of insider trading and analyst recommendations are more likely to be complementary.

This research offers several contributions to the literature. First, we argue and provide novel evidence that one group of informed agents does not disincentivize the other group from disseminating private information to the market. Instead, both could provide valuable information to the market due to their disparate motives and constraints. Second, to the best of our knowledge, our study is the first to examine the joint and conditional informativeness of insider trading and analyst recommendations. It not only affords a rare opportunity to compare the signals contained in these two information-revealing activities directly but also provides significant insights into their informational roles. Third, our research employs a unique approach to assessing the investment value of analyst stock recommendations. Notably, since prior studies typically evaluate analyst recommendations by drawing inferences from their associated return patterns, their findings are prone to several drawbacks related to portfolio formation strategies in the research design, stock characteristics, and the studied sample period, inevitably resulting in mixed evidence.[3] Our approach circumvents such shortcomings by examining how one group of market participants – corporate insiders – trade on their firm's stock vis-*â*-vis the recommendations issued by analysts. The direction and magnitude of insider trades, relative to analyst recommendations, reveal insiders' explicit responses to the investment value of their firm's stock assessed by analysts. Fourth, our results shed light on the long-raging controversies on insider trading regulations and the securities analyst industry. For instance, proponents of insider trading regulations argue that constraining insider trading promotes market fairness and increases liquidity. Critics of insider trading regulations, however, suggest that permitting insiders to benefit from their informational advantage in their trading improves price efficiency (e.g., Leland, 1992; George and Seyhun, 2002). Both arguments imply that insiders and other information intermediaries such as analysts substitute for each other's informational role in the market. As a result, active insider trading could disincentivize analysts from engaging in information-production activities. Our findings mostly support the view that insiders and analysts complement each other in providing value-pertinent information to the market.

This paper is organized as follows. Section 2 summarizes our hypotheses. Section 3 describes the databases employed in this study. Section 4 investigates the interactions between insiders and financial analysts. Section 5 evaluates the return predictability and the relative informativeness of insider trading and analyst recommendations. Section 6 conducts additional robustness tests, and the final section concludes.

2. Hypotheses

We first summarize the hypotheses related to the interactions between insiders and analysts, followed by those related to the informativeness of insider trading and analyst recommendations.

2.1. Interactions between insiders and analysts

2.1.1. Effect of insider trading on analyst recommendations

H1A. The Information Hypothesis *: The directions of insider trading and subsequent analyst recommendation revisions are the same.*

H1B. The Signaling Hypothesis *: The directions of insider trading and subsequent analyst recommendation revisions are the opposite.*

H1C. The Biased Information Processing Hypothesis *: Insider trading and subsequent analyst recommendation revisions are uncorrelated.*

[3] Early research finds no evidence of superior investment performance (e.g., Bidwell, 1977; Groth et al., 1979). Several recent studies show that, on average, stock recommendations are at least short-term informative, while others argue that only a small set of stock recommendations have investment value (e.g., Dimson and Marsh, 1984; Elton et al., 1986; Stickel, 1995; Womack, 1996; Barber et al., 2001; Altinkilic and Hansen, 2009; Loh and Stulz, 2011; Li et al., 2015). In addition, Jegadeesh et al. (2004) show that the change in recommendations has more value than the level of recommendations.

2.1.2. Effect of analyst recommendations on insider trading

H2A. The Information Hypothesis *: The directions of analyst recommendation revisions and subsequent insider trading are the same.*

H2B. The Signaling Hypothesis *: The directions of analyst recommendation revisions and subsequent insider trading are the opposite. This relationship should hold mainly due to the effect of analyst downgrades on insider purchases.*

H2C. The Liquidity Hypothesis *: Following analyst upgrades, insiders are more likely to sell their share holdings for reasons unrelated to information.*

2.2. Conditional and joint informativeness of insider trades and analyst recommendations

A critical premise in determining the conditional and joint informativeness of insider trading and analyst recommendations is whether their activities *alone* are valuable to other investors. The null hypothesis is that neither activity is informative if insider trading and analyst recommendations are simply responses to common public information and reveal little new information to the market. While, under this null, both activities could exhibit patterns consistent with the hypotheses laid out in the previous sections, any observed price reactions following their actions should be minimal. Importantly, as discussed above, even without new information, insiders could opportunistically induce mispricing of their stock before trading through manipulating earnings or disclosing material news (Cheng and Lo, 2006; McVay et al., 2006; Sawicki and Shrestha, 2008; Korczak et al., 2010). However, the information effect of such insider trading is likely transitory rather than permanent, and share prices should reverse subsequently.

Alternatively, if the signals from at least one group are consequential, significant stock price movements should follow. If both signals are informative, their distinction and relative strength will dictate the direction of future stock price movements. Using the discussions in the Introduction section, we summarize the following two hypotheses related to market reactions to the signals from these two groups of informed agents.

2.2.1. The substitution hypothesis

H3 *: If one of the two signals (insider trading & analyst recommendations) occurs first, the subsequent signal is rendered uninformative.* Hence, conditional on insider trading, the subsequent abnormal return associated with an analyst upgrade or downgrade should be similar; i.e., $RET(Upgrade_T|Insider\ Buy_{T-1})=RET(Downgrade_T|Insider\ Buy_{T-1})$ and $RET(Upgrade_T|Insider\ Sell_{T-1})=RET(Downgrade_T|Insider\ Sell_{T-1})$. Similarly, conditional on analyst recommendation revisions, there is no significant difference in abnormal returns associated with insider buying vs. selling; i.e., $RET(Insider\ Buy_T|Upgrade_{T-1})=RET(Insider\ Sell_T|Upgrade_{T-1})$ and $RET(Insider\ Buy_T|Downgrade_{T-1})=RET(Insider\ Sell_T|Downgrade_{T-1})$. The above equations also imply that no matter whether the two signals concur or disagree, their joint market reactions should follow the dominating individual signal. That is, for downgrades, $RET(Insider\ Sell \cap Downgrade) = [RET(Insider\ Sell)$ or $RET(Downgrade)]$ and $RET(Insider\ Buy \cap Downgrade)= [RET(Insider\ Buy)$ or $RET(Downgrade)]$, while for upgrades, $RET(Insider\ Sell \cap Upgrade) = [RET(Insider\ Sell)$ or $RET(Upgrade)]$ and $RET(Insider\ Buy \cap Upgrade)= [RET(Insider\ Buy)$ or $RET(Upgrade)]$, depending on which signal dominates.

2.2.2. The complementarity hypothesis

H4 *: If one of the two signals (insider trading & analyst recommendations) occurs first, the subsequent signal continues to be informative.* Thus, conditional on insider trading, $RET(Upgrade_T|Insider\ Buy_{T-1}) > RET(Downgrade_T|Insider\ Buy_{T-1})$ and $RET(Upgrade_T|Insider\ Sell_{T-1}) > RET(Downgrade_T|Insider\ Sell_{T-1})$, while conditional on analyst recommendations, $RET(Insider\ Buy_T|Upgrade_{T-1}) > RET(Insider\ Sell_T|Upgrade_{T-1})$ and $RET(Insider\ Buy_T|Downgrade_{T-1}) > RET(Insider\ Sell_T|Downgrade_{T-1})$. In addition, the complementarity hypothesis implies that if both signals are effective, then when they concur, their joint market reaction should be stronger than the individual market reaction of each of the signals, whereas when they disagree, then their joint market reaction should be between the market reactions of negative and positive individual signals as the reactions to two contrasting signals will be attenuated. That is, if the two signals concur, then for downgrades, $RET(Insider\ Sell \cap Downgrade) < RET(Insider\ Sell)$ and $RET(Insider\ Sell \cap Downgrade) < RET(Downgrade)$, whereas if they disagree, then $RET(Downgrade) < RET(Insider\ Buy \cap Downgrade) < RET(Insider\ Buy)$. For upgrades, $RET(Insider\ Buy \cap Upgrade) > RET(Insider\ Buy)$, $RET(Insider\ Buy \cap Upgrade) > RET(Upgrade)$, and $RET(Insider\ Sell) < RET(Insider\ Sell \cap Upgrade) < RET(Upgrade)$.

3. Data and sample construction

Our study employs data from four major sources covering the sample period from 1994 to 2016: (1) earnings announcement dates and consensus analyst recommendations from I/B/E/S, (2) insider transactions and institutional holdings from Thomson Reuters, (3) accounting information from Compustat, and (4) stock returns from CRSP. We only include common stocks with share codes 10 or 11 and with available equity and accounting data.

3.1. Analyst recommendations

We utilize I/B/E/S Summary Statistics file that contains the consensus recommendations for each stock covered by at least one analyst.[4] It is important to note that we focus on the interactions between insiders and analysts at the firm level. Thus, in contrast to most studies investigating the information content of individual analyst recommendations, we resort to consensus recommendations and study analyst recommendations as a whole for each firm. One advantage of using consensus recommendations is that it captures the overall 'net' analyst opinion about the stock during a month. One potential drawback, however, is that all recommendations are weighted equally; hence, it does not consider the evidence that some individual recommendations are more influential than others (see, e.g., Loh and Stulz, 2011). Analyst consensus recommendations are reported as a numerical value between 1 and 5, inclusive, with 1 indicating a Strong Buy recommendation and 5 a Strong Sell. To ease interpretation, we reverse the reported rankings by using the following formula: New Consensus Recommendation = 6 - Reported Consensus Recommendation. After conversion, the recommendation ratings have a more intuitive interpretation: a higher rating or an increase in rating indicates a more favorable consensus opinion or an upgrade from financial analysts.

Columns [1]-[10] in Panel A of Table 1 provide a snapshot of the sample, which on average, includes 4,124 firms (70.67% of the CRSP universe) covered by at least one analyst per year. The average consensus rating is fairly high at 3.79, and the average percentage of buy or strong buy recommendations, collectively reported as "% of Buys", is 57.58%. Further, the mean and median numbers of analysts covering a firm are 6.81 and 4.61, respectively, although the remaining statistics indicate a wide range of the number of analysts per firm.[5]

[4] I/B/E/S calculates the consensus recommendation for each stock on the third Thursday of each month based on all outstanding recommendations for the stock. Inactive or stopped recommendations are excluded from the calculation.

[5] Appendix Table A.1 presents the sample distributions by year. Interestingly, the overall pattern shifts after two important regulatory changes: Regulation Fair Disclosure (RegFD) in 2000 and Global Analyst Research Settlement (GARS) in 2002.

Table 1

Descriptive statistics of analyst coverage, insider trades, and stock characteristics. This table presents the average values of summary statistics on the number of firms with analyst coverage, the number of analysts per firm, and aggregate insider trades (Panel A) as well as stock characteristics (Panel B). The sample of recommendations are from I/B/E/S Consensus Recommendations US file, and insider transactions come from Form 4, which corporate insiders file with the SEC when their ownership changes. In Panel B, the 'Downgrade' ('Upgrade') portfolio consists of stocks that have been downgraded (upgraded) during the month, and the 'No Change' portfolio consists of stocks whose consensus recommendations remain unchanged. Market value of equity (MV, $million) is price per share times total shares outstanding and BM is the book-to-market equity ratio at the end of previous fiscal year. MOM is the firm's past twelve-month characteristics-adjusted return. Volatility is the average daily stock return volatility in the past month. Turnover is the average daily trading volume divided by total shares outstanding in the past month. Institutional holdings is normalized by the total number of shares outstanding. % Top 10 Brokers is the percentage of analyst recommendations from the top ten brokers for each firm in a month.

Panel A. Analyst Coverage and Aggregate Insider Trades

Firms with Analyst Coverage					No. of Analyst Coverage per Firm					Aggregate Insider Transactions					
No. of	No. of	% of								$Trades (B)		#Trades (000)		% of Net Trades	
Firms with Coverage	Firms on CRSP	Firms with Coverage	Mean Rating	% of Buys	Mean	Median	Min	Max	Std	Buys (i)	Sells (ii)	Buys (iii)	Sells (iv)	PVB = $\frac{(i)-(ii)}{(i)+(ii)}$	PNB = $\frac{(iii)-(iv)}{(iii)+(iv)}$
[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	[11]	[12]	[13]	[14]	[15]	[16]
4124	5839	70.67	3.79	57.58	6.81	4.61	1	55	6.41	0.83	13.35	10.02	41.38	-87.85	-55.00

Panel B. Stock Characteristics

Coverage/Consensus Recommendation Revisions	N	Mean Consensus Recommendation	MV ($MM)	BM	MOM	Volatility	Turnover	Institutional Holdings	% Rec from Top 10 Brokers
	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
Least Coverage	**162,183**	**3.83**	**252.79**	**0.8878**	**0.0925**	**0.0381**	**0.0032**	**0.2788**	**0.0006**
Downgrade	9,418	3.58	248.24	1.0418	3.2337	0.0404	0.0046	0.2878	0.0008
No Change	147,021	3.83	222.34	0.8287	3.0868	0.0379	0.0031	0.2562	0.0006
Upgrade	5,744	4.33	294.27	0.7417	3.1422	0.0380	0.0047	0.2934	0.0004
Quintile 2	**203,297**	**3.86**	**515.47**	**0.7497**	**0.1204**	**0.0356**	**0.0042**	**0.4086**	**0.0027**
Downgrade	22,638	3.67	532.65	0.9026	3.1521	0.0385	0.0056	0.3965	0.0032
No Change	162,566	3.86	601.58	0.7161	3.0430	0.0352	0.0039	0.4254	0.0027
Upgrade	18,093	4.10	402.31	0.6114	3.0976	0.0361	0.0053	0.4039	0.0022
Quintile 3	**209,889**	**3.84**	**715.49**	**0.6500**	**0.1485**	**0.0327**	**0.0054**	**0.5207**	**0.0061**
Downgrade	36,000	3.71	686.44	0.7905	3.0481	0.0354	0.0067	0.4990	0.0066
No Change	143,093	3.85	774.84	0.6320	2.9845	0.0320	0.0049	0.5357	0.0067
Upgrade	30,796	3.96	684.81	0.5201	3.0338	0.0330	0.0064	0.5273	0.0050
Quintile 4	**218,733**	**3.78**	**1,694.47**	**0.5991**	**0.1633**	**0.0292**	**0.0072**	**0.6248**	**0.0129**
Downgrade	53,464	3.69	1,559.09	0.7420	2.9456	0.0318	0.0086	0.5768	0.0133
No Change	117,441	3.79	1,839.14	0.5730	2.9304	0.0279	0.0063	0.6421	0.0141
Upgrade	47,828	3.85	1,684.84	0.4799	2.9549	0.0296	0.0080	0.6555	0.0111
Most Coverage	**222,788**	**3.72**	**13,277.52**	**0.5472**	**0.1448**	**0.0249**	**0.0089**	**0.6858**	**0.0237**
Downgrade	77,147	3.67	8,275.80	0.6675	2.7963	0.0267	0.0097	0.6362	0.0249
No Change	73,013	3.72	14,180.24	0.5155	2.8424	0.0230	0.0076	0.6940	0.0246
Upgrade	72,628	3.77	17,339.43	0.4594	2.8538	0.0250	0.0092	0.7269	0.0217

3.2. Insider transactions

Insider transactions are obtained from Form 4 that corporate insiders file with the Securities and Exchange Commission (SEC) when their ownership changes. The information includes the identity and classification of the insider, along with the date, type, price, and the number of shares for each transaction. We define insiders as high-level officers and directors as trading gains exist primarily in this upper management level. In addition, since we focus on the informational role of insider trades, our analysis is restricted to open-market transactions of equity shares. It thus excludes transactions related to employee stock option exercises, which are likely motivated by liquidity and portfolio diversification considerations rather than by private information (Ofek and Yermack, 2000; Carpenter and Remmers, 2001). We also delete duplicate records reported by insiders, trading records with no transaction prices, and all odd-lot transactions (fewer than 100 shares).

We utilize several variables to measure insider trading activities, and all are estimated monthly to align with the monthly consensus recommendations.[6] The first two measures, 'Insider Buy' ($Buys) and 'Insider Sell' ($Sells), are the market values of insider purchases and sales, respectively, scaled by the market value of all insider trades. The next two measures (also Insider Buy as #Buys and Insider Sell as #Sells) use their share numbers rather than market values. We use these four measures to examine whether insider buys versus sales mainly drive our results and insiders' motives therein. The fifth, PVB, is the market value of 'Net Insider Trade'; i.e., PVB=($Buys - $Sells)/($Buys + $Sells). PVB equals zero if there is no insider trade during a month. The final measure, PNB, is based on the share number; i.e., PNB=(#Buys - #Sells)/(#Buys + #Sells).[7] Both PVB and PNB are normalized measures of insiders' *net* purchases, and more importantly, capture the relative signal strength of insider purchases vs. sales across firms with varying degrees of insider trading.

Columns [11]-[16] in Panel A provide the average market value and number of insider buys and sales, as well as PVB and PNB, per year. Consistent with prior literature, we find that the value of insider sales far outweighs the value of buys. The average insider purchase amount is $0.83 billion, which pales in comparison with the sale amount of $13.35 billion. The number of insider purchases is also smaller than the number of insider sales, 10,020 vs. 41,380 trades. The resulting average PVB and PNB are -87.55% and -55.00%, respectively.[8]

3.3. Control variables

Our multivariate analyses include several stock and analyst characteristics that could affect analyst recommendations and/or insider trades (e.g., Rozeff and Zaman, 1998; Lakonishok and Lee, 2001; Jegadeesh et al., 2004; Jenter, 2005), namely past monthly return, firm size (MV), book-to-market (BM), momentum (MOM), volatility, turnover, analyst coverage, institutional holdings, and the percentage of recommendations from top ten brokers. Note that we include both past monthly return and MOM to control for not only the tendencies that insiders are contrarians while analysts prefer momentum but also the possibility that insiders could strategically push share prices lower (higher) by manipulating earnings downward (upward) or disclosing bad (good) news before purchasing (selling) shares.[9] In addition, we include institutional holdings to control for the quality of a firm's corporate governance environment (e.g., Shleifer and Vishny, 1986; Hartzell and Starks, 2003) and the percentage of analyst recommendations from *Institutional Investor's* Top 10 brokerages for each firm to control for potential conflicts of interest between an investment bank's corporate finance business and its analyst research (e.g., Michaely and Womack, 1999; Ljungqvist et al., 2007).

Panel B of Table 1 summarizes the stock-level mean consensus recommendations and stock characteristics across the 15 double-sorted portfolios. Specifically, we sort stocks per year into quintiles based on the average number of analysts covering the stocks. All stocks covered in the I/B/E/S database are included in the first sort irrespective of the availability of insider trading data to ensure that the stock ranking is not conditional on insider trading and hence not biased in a discernible way. The bottom and top quintiles ('Least Coverage' & 'Most Coverage') contain stocks with the fewest and most analysts following, respectively. Then, within each coverage quintile, we further sort stocks into terciles based on analyst recommendation revisions. The '"Downgrade"' ('Upgrade') tercile consists of stocks whose ratings have fallen (risen) during a month while the middle '"No Change"' tercile contains stocks that experience no rating changes. Note that the purpose of the first sort is to highlight the importance of controlling for the impact of analyst coverage on consensus recommendations and certain stock characteristics in the univariate analysis. Without the first sort, some portfolios will disproportionately contain a large number of low-coverage (and likely small) stocks since firms with only one or two analysts following tend to have more extreme ratings and rating changes than those with a greater number of analysts. Moreover, it is documented that both analyst coverage and insider trading are correlated with firm size (e.g., Bushan, 1989; Womack, 1996; Lakonishok and Lee, 2001). We include both firm size and analyst coverage as controls in the multivariate analysis.

Column [1] reveals that analysts do not revise their recommendations frequently, as evident in the No Change portfolio having the largest number of observations except for the Most Coverage quintile. Even though analysts might update their forecasts often, they typically do not revise their recommendation ratings unless they believe the stock deserves a different one. Although the mean consensus ratings remain fairly stable across coverage quintiles (as shown in column [2]), their spreads between the Downgrade and Upgrade portfolios vary significantly across the coverage levels, and firms with the least (most) analyst coverage have the largest (smallest) rating spreads: 0.75 versus 0.10, thereby indicating the impact of analyst coverage on its consensus rating changes and the importance of controlling for such effect in the analysis.

The remaining columns report the mean values of the control variables within each portfolio.[10] As expected, firms with lower an-

For example, the mean analyst consensus ratings prior to 2002 are all higher than those after 2002. In addition, the average percentage of firms receiving a buy or higher recommendation peaked at 69.58% in 2000 but declined to the lowest point at 46.51% in 2003. Since brokerage firms are likely to experience structural changes following these two regulatory events, we conduct robustness tests in a later section using a subsample of observations post-RegFD and GARS (i.e., post-2002).

[6] One potential concern is that since insiders do not trade frequently, and analysts do not consistently update their recommendations, measuring both variables monthly might not detect useful information. This concern, if valid, will run against us finding significant results. In earlier versions of the paper, we also estimate and report measures at the quarterly frequency, the results of which, available upon request, are qualitatively similar to the monthly results.

[7] Existing studies employ variations of the above measures as proxies for insider trading activity (e.g., Rozeff and Zaman, 1998; Lakonishok and Lee, 2001). Following those studies, we use different measures to ensure the robustness of our results. We report and discuss results in several tables using the PVB measure and relegate the PNB results to the Appendix.

[8] The time-series distributions of these variables are presented in Appendix Table A.1.

[9] See, e.g., Cheng and Lo (2006); Sawicki and Shrestha (2008), and Korczak et al. (2010). Although it is plausible that insiders use tactics to increase their trading profits, Bettis et al. (2000) document that the majority of their sample firms have their policies to restrict trading by insiders.

[10] For MV, BM, and MOM, we follow Daniel and Titman's (1997) procedure to assign rankings. Specifically, at the end of June each year, we first sort all stocks

alyst coverage are generally smaller, suggesting that analyst coverage provides a reasonably good proxy for firm size. They are also associated with higher BM ratios and return volatility, but with lower MOM, turnover, institutional holdings, and percentage of recommendations from top ten brokers. Further, within each coverage portfolio, the Downgrade portfolio has a higher mean BM than the Upgrade portfolio while both portfolios usually generate higher mean MOM ranking than the No Change portfolio, indicating that the analyst recommendations for stocks with higher past stock returns are likely to be revised. The overall patterns in the control variables emphasize the importance of controlling for the effect of analyst coverage when analyzing analyst opinions and insider trading activities.

4. Dynamics between insider trading and analyst recommendations

In this section, we test the hypotheses related to whether and how the actions of one informed group subsequently affect those of the other. These intertemporal tests offer clues about the interactions, or the lack thereof, between insiders and analysts.

4.1. Effects of insider trading on analyst recommendations

We first examine whether analysts take cues from insiders when revising their recommendations. For an univariate analysis, we calculate the average net insider trade (PVB and PNB) in the month prior to analyst recommendation revisions for the three portfolios: Downgrade, No Change, and Upgrade. If insider trading affects subsequent analyst recommendation revisions, we should observe systematic differences in net insider trades between upgrades and downgrades; otherwise, differences should be insignificant. We further conduct our tests using a subsample of top insiders that include the CEO, President, CFO, General Counsel, and Chairman of the Board. They arguably possess more timely private information although they are also subject to closer media scrutiny and more prone to insider trading regulations, reducing their incentives to trade and potentially making their trades less informative than other insiders.

The results reported in Panels A.1 and A.2 of Table 2 reveal little evidence that analysts take cues from insider trading when revising their recommendations. Both ratio differences (Dn - Up) are almost zero and statistically insignificant across the bottom four coverage quintiles. The ratio differences for the Most Coverage quintile are significantly positive, suggesting that when insiders signal positive news with more purchases, analysts are likely to downgrade, rather than upgrade, the stock in the subsequent month, and this contradicting pattern persists in both samples.

Next, we conduct the test using the following specification:

$$REC_{it} = \alpha + \lambda_j + \tau_t + \delta(Insider\ Trading_{it-1}) + \sum_{k}^{K} \beta_k X_{kit-1} + \epsilon_{it}, \tag{1}$$

where *REC* is analyst recommendation change (Rec Change), Upgrade, or Downgrade, *Insider Trading* is Net Insider Trade (PVB or PNB), Insider Buy, or Insider Sell, λ_j and τ_t are industry and year fixed effects, respectively, and X_{kit-1} is the vector of control variables described earlier. Rec Change is a numerical change in consensus analyst recommendations from previous to current months while Upgrade and Downgrade are indicator variables that capture positive and negative recommendation changes, respectively.

Table 3 presents the estimation results using PVB.[11] The dependent variable is Rec Change in columns (1)-(2), Upgrade in (3)-(4) and Downgrade in the last two, where results in columns (3)-(6) are estimated using probit regressions. The multivariate results indicate that the impact of insider trading on analyst recommendation revisions is somewhat mixed after controlling for other factors. The first three models in Panel A show that analysts are more likely to upgrade the stocks they cover following insider (net) purchases, but the economic magnitude in model 1 is small. Models 4 and 6 reveal that the probability of an analyst upgrade (downgrade) is lower following insider selling (buying). However, we also find a mixed role of insider selling as Model 6 further indicates that analysts are also less likely to downgrade the stock following insider selling. This implies that insider selling is less informative than buying or that analysts tend to employ other information than insider trading when making downgrade decisions. Finally, Panel B shows weaker results, suggesting that analysts probably consider trades from top insiders less informative as their trades are subject to more intense scrutiny than other insiders'.

4.2. Effects of analyst recommendations on insider trading

We next examine whether insiders take cues from analyst recommendations when making their trading decisions and evaluate the motives behind their trades. Again, we begin by conducting a simple univariate test by using Net Insider Trade (PVB and PNB). As shown in Panels B.1 (full sample) and B.2 (top insiders only) of Table 2, insiders, surprisingly, trade in the opposite directions of analyst recommendation revisions; they have more purchases than sales following downgrades, but the pattern reverses following upgrades. Both panels indicate consistently higher net buy ratios for Downgrade than Upgrade portfolios, resulting in significantly positive ratio differentials (Dn - Up) across all coverage quintiles.

This contrasting pattern of insider trading vis-*à*-vis analyst recommendation revisions seems inconsistent with the information hypothesis as analysts' actions do not significantly deter insiders from trading, and yet they trade differently from analyst recommendations. We, therefore, explore the remaining two possible explanations, namely signaling and liquidity-driven, using the following specification:

$$Insider\ Trading_{it} = \alpha + \lambda_j + \tau_t + \delta REC_{it-1} + \sum_{k}^{K} \beta_k X_{kit-1} + \epsilon_{it}. \tag{2}$$

The results using PVB for *Insider Trading* are reported in Panels A (full sample) and B (top insiders only) of Table 4.[12] Consistent with the univariate evidence, the regression results in models 1 and 2 of both panels show that analyst recommendation revisions significantly influence insider trading, but insiders trade against the directions of those revisions. When analysts upgrade the stocks, insiders have more sales than buys whereas insiders reverse their trades following analyst downgrades. The former is consistent with the liquidity hypothesis while the latter is in line with the signaling hypothesis. The significantly negative impact of analyst recommendation revisions on insider trading persists across both samples even after controlling for insiders' preference for value stocks (as shown in positive Book-to-Market estimates)

into size quintiles using NYSE market capitalization breakpoints. Stocks in each size portfolio are then sorted into quintiles based on their BM ratios. Finally, the stocks in each of the 25 size/BM portfolios are sorted into quintiles based on their prior 12-month returns. We then assign each stock in our sample into one of the 125 portfolios. A ranking of "1" ('5') is assigned to stocks in the lowest (highest) size, BM, or MOM quintile. The mean ranking of each coverage portfolio is the average of all rankings of stocks in that portfolio.

[11] The results using PNB are qualitatively similar and reported in Panel A of Appendix Table A.2.

[12] The results using PNB are reported in Panel B of Appendix Table A.2

Table 2

Univariate analysis of insider trades and analyst recommendation revisions. The table presents insider trading activity in the preceding (Panels A.1. and A.2.) and subsequent (Panels B.1. and B.2.) months as recommendation revisions. Insider trading activity is the average monthly net insider trades defined as either the percentage of the *value* of insider buys minus sales to the value of all insider trades ('PVB') or the percentage of the *number* of insider buys minus sales to the number of all insiders' trades ('PNB'). If there is no insider trade within a month, both PVB and PNB are set to zero. The 'Downgrade' ('Upgrade') portfolio consists of stocks that have been downgraded (upgraded) from the previous month, and the 'No-Change' portfolio consists of stocks whose consensus recommendations remain unchanged. For each panel, we present two different sets of results. The first set utilizes the full sample while the second only includes top insiders who are CEO, President, CFO, General Counsel, and Chairman of the Board. The significance level of the differences in insider trading activity between the Downgrade and Upgrade portfolios ('Dn-Up') is based on a two-sided t-test, and associated $t-$statistics are reported in parentheses. (***), (**), and (*) indicate significance at the 1%, 5%, and 10% levels, respectively.

	Least Coverage			Quintile 2			Quintile 3			Quintile 4			Most Coverage		
	N	PVB	PNB	N	PVB	PNB	N	PVB	PNB	N	PVB	PNB	N	PVB	PNB
Panel A.1. Insider Trading Activity and *Subsequent*Analyst Recommendation Revisions															
Downgrade (Dn)	6,084	-0.003	-0.001	20,080	-0.034	-0.031	33,969	-0.069	-0.065	52,822	-0.107	-0.103	76,992	-0.135	-0.130
No Change	147,014	0.014	0.016	162,566	-0.021	-0.018	143,093	-0.065	-0.061	117,441	-0.102	-0.098	73,013	-0.143	-0.138
Upgrade (Up)	5,742	-0.003	-0.001	18,093	-0.027	-0.026	30,796	-0.072	-0.069	47,828	-0.104	-0.101	72,625	-0.142	-0.138
Dn - Up		-0.000	0.000		-0.007	-0.006		0.003	0.004		-0.002	-0.003		0.008***	0.007***
(t-stat.)		(-0.05)	(0.02)		(-1.60)	(-1.41)		(0.98)	(1.12)		(-0.79)	(-0.88)		(3.19)	(3.02)
Panel A.2. Top Insider Trading Activity and *Subsequent*Analyst Recommendation Revisions															
Downgrade (Dn)		0.002	0.002		-0.011	-0.010		-0.033	-0.032		-0.051	-0.051		-0.058	-0.058
No Change		0.010	0.010		-0.005	-0.005		-0.028	-0.028		-0.047	-0.046		-0.060	-0.060
Upgrade (Up)		0.005	0.005		-0.007	-0.007		-0.034	-0.034		-0.051	-0.051		-0.062	-0.062
(Dn-Up)		-0.003	-0.002		-0.003	-0.003		0.002	0.002		-0.001	-0.001		0.004***	0.004***
		(-0.56)	(-0.53)		(-1.16)	(-1.11)		(0.77)	(0.83)		(-0.33)	(-0.31)		(2.79)	(2.79)
Panel B.1. Analyst Recommendation Revisions and *Subsequent*Insider Trading Activity															
Downgrade (Dn)	9,312	0.012	0.014	22,380	-0.021	-0.018	35,626	-0.059	-0.056	53,522	-0.096	-0.093	77,302	-0.133	-0.129
No Change	143,812	0.012	0.015	160,505	-0.023	-0.020	141,741	-0.068	-0.064	116,840	-0.101	-0.097	72,523	-0.136	-0.132
Upgrade (Up)	5,716	-0.012	-0.008	17,854	-0.041	-0.037	30,491	-0.081	-0.078	47,729	-0.120	-0.116	72,805	-0.150	-0.146
Dn - Up		0.023***	0.022***		0.020***	0.019***		0.022***	0.023***		0.025***	0.024***		0.017***	0.017***
(t-stat.)		(3.59)	(3.44)		(4.93)	(4.72)		(6.53)	(6.67)		(8.47)	(8.27)		(6.93)	(7.01)
Panel B.2. Analyst Recommendation Revisions and *Subsequent*Top Insider Trading Activity															
Downgrade (Dn)		0.010	0.011		-0.005	-0.004		-0.027	-0.027		-0.046	-0.045		-0.057	-0.057
No Change		0.009	0.010		-0.006	-0.006		-0.030	-0.030		-0.046	-0.046		-0.059	-0.058
Upgrade (Up)		-0.001	-0.001		-0.014	-0.014		-0.037	-0.037		-0.057	-0.057		-0.065	-0.064
(Dn-Up)		0.012***	0.011***		0.010***	0.010***		0.010***	0.010***		0.011***	0.011***		0.007***	0.007***
		(2.69)	(2.65)		(3.63)	(3.75)		(4.45)	(4.59)		(5.85)	(5.90)		(4.54)	(4.65)

Table 3

Effects of insider trading on subsequent analyst recommendation revisions. The table shows results from panel and logistic regressions of analyst recommendation revisions in the subsequent month. The dependent variable is recommendation change (models 1 and 2) or the probability of upgrade (3 and 4) or downgrade (5 and 6). Insider Buy (Insider Sell) is defined as the ratio of the value of insider purchases (insider sales) to the value of all insider trades. Net Insider Trade is Insider Buy minus Insider Sell. The regression results are estimated using the full sample (Panel A) and the subsample of top insiders only (Panel B). In the subsample, insider trading activity is estimated only from top insiders who are CEO, President, CFO, General Counsel, and Chairman of the Board. All explanatory variables are lagged variables. The analyst recommendations are re-coded as follows: a rating of '1' reflects a strong sell recommendation, '2' a sell, '3' a hold, '4' a buy, and '5' a strong buy recommendation. Recommendation (Rec) Change is consensus analyst recommendation change from month T-1 to month T. Upgrade (Downgrade) is an indicator variable if Recommendation Change is positive (negative). Previous month return is the characteristics-adjusted abnormal return in month T-1. MOM is the firm's past twelve-month characteristics-adjusted abnormal return. Market value of equity (MV) is price per share times total shares outstanding and the book-to-market ratio is measured at the end of previous fiscal year. Volatility is the average daily stock return volatility in the past month. Turnover is the average daily trading volume divided by total shares outstanding in the past month. Institutional holdings is normalized by the total number of shares outstanding. Analyst Coverage is a ranking variable (1 to 5) based on the number of analysts covering the stock in the previous year. Institutional holdings are institutional ownership measured in the previous quarter. % Top 10 Brokers is the percentage of analyst recommendations from the top ten brokers for each firm in a month. Industry fixed effects (FE) are based on Fama-French 30-industry groupings, and year FEs are yearly indicator variables during the sample period except for year 2016 to ensure correct model specifications. $T-$statistics, reported in parentheses, are associated with standard errors adjusted for firm-level clustering. (***), (**), and (*) indicate significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Full Sample

	Rec Change		Upgrade		Downgrade	
	[1]	[2]	[3]	[4]	[5]	[6]
Net Insider Trade	0.001*		0.021***		0.009	
	(1.68)		(3.23)		(1.41)	
Insider Buy		0.003***		-0.007		-0.061***
		(3.64)		(-0.54)		(-5.00)
Insider Sell		0.001		-0.035***		-0.042***
		(1.06)		(-4.20)		(-5.11)
Previous Month Return	0.003	0.003	-0.001	-0.002	0.034	0.032
	(1.50)	(1.52)	(-0.03)	(-0.07)	(1.58)	(1.49)
Momentum	0.010***	0.010***	0.099***	0.100***	-0.065***	-0.064***
	(21.77)	(21.73)	(17.73)	(17.73)	(-11.05)	(-10.96)
log(MV)	0.002***	0.002***	0.085***	0.086***	0.052***	0.052***
	(13.90)	(13.78)	(29.04)	(29.10)	(18.39)	(18.54)
log(Book-to-Market)	0.005***	0.005***	0.004	0.004	-0.075***	-0.076***
	(20.63)	(20.71)	(0.96)	(0.89)	(-17.22)	(-17.38)
Volatility	-0.128***	-0.128***	2.367***	2.371***	4.858***	4.868***
	(-7.96)	(-7.97)	(13.30)	(13.32)	(25.80)	(25.83)
Turnover	-0.646***	-0.647***	4.929***	4.937***	11.424***	11.446***
	(-13.55)	(-13.56)	(10.15)	(10.15)	(17.64)	(17.63)
Analyst Coverage	-0.001***	-0.001***	0.463***	0.463***	0.486***	0.486***
	(-6.29)	(-6.31)	(112.76)	(112.72)	(121.80)	(121.68)
Institutional Holdings	-0.006***	-0.006***	0.078***	0.078***	0.162***	0.162***
	(-9.64)	(-9.70)	(5.36)	(5.38)	(11.12)	(11.18)
% Top 10 Brokers	-0.016***	-0.016***	0.306***	0.305***	0.470***	0.469***
	(-7.03)	(-7.00)	(10.56)	(10.58)	(17.49)	(17.44)
Intercept	0.003*	0.002	-4.169***	-4.167***	-4.014***	-4.009***
	(1.70)	(1.58)	(-105.38)	(-105.23)	(-109.10)	(-108.60)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	880,897	880,897	880,897	880,897	880,897	880,897
R^2	0.003	0.003				

Panel B. Top Insiders Only

	Rec Change		Upgrade		Downgrade	
	[1]	[2]	[3]	[4]	[5]	[6]
Net Insider Trade	0.001		0.011		0.001	
	(1.39)		(1.19)		(0.13)	
Insider Buy		0.003**		0.014		-0.068***
		(2.33)		(0.66)		(-3.57)
Insider Sell		0.000		-0.011		-0.027**
		(0.20)		(-0.96)		(-2.23)
Previous Month Return	0.003	0.003	-0.004	-0.004	0.032	0.031
	(1.48)	(1.49)	(-0.18)	(-0.18)	(1.49)	(1.44)
Momentum	0.010***	0.010***	0.099***	0.099***	-0.065***	-0.065***
	(21.77)	(21.74)	(17.68)	(17.67)	(-11.09)	(-11.04)
log(MV)	0.002***	0.002***	0.085***	0.085***	0.052***	0.051***
	(13.86)	(13.87)	(28.98)	(28.97)	(18.36)	(18.31)
log(Book-to-Market)	0.005***	0.005***	0.005	0.005	-0.075***	-0.075***
	(20.63)	(20.67)	(1.06)	(1.06)	(-17.17)	(-17.24)
Volatility	-0.128***	-0.128***	2.383***	2.383***	4.867***	4.880***
	(-7.94)	(-7.97)	(13.39)	(13.39)	(25.85)	(25.88)
Turnover	-0.646***	-0.647***	4.935***	4.935***	11.427***	11.448***
	(-13.55)	(-13.56)	(10.16)	(10.16)	(17.65)	(17.65)
Analyst Coverage	-0.001***	-0.001***	0.463***	0.463***	0.486***	0.486***
	(-6.30)	(-6.31)	(112.76)	(112.76)	(121.77)	(121.76)
Institutional Holdings	-0.006***	-0.006***	0.076***	0.076***	0.161***	0.162***
	(-9.69)	(-9.74)	(5.27)	(5.27)	(11.08)	(11.14)
% Top 10 Brokers	-0.016***	-0.016***	0.306***	0.306***	0.470***	0.471***
	(-7.03)	(-7.03)	(10.56)	(10.56)	(17.49)	(17.49)
Intercept	0.003*	0.003*	-4.168***	-4.168***	-4.014***	-4.010***
	(1.73)	(1.65)	(-105.43)	(-105.46)	(-109.14)	(-108.83)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	880,897	880,897	880,897	880,897	880,897	880,897
R^2	0.003	0.003				

and their contrarian trading (as shown in negative Previous Month Return and Momentum estimates). Thus, the markedly negative effect is not only inconsistent with the information argument, but also more than driven by insiders' taking advantage of mispricing.

To provide further insights into these findings, we next include additional variables in the models and run separate regressions for insider purchases versus sales. Specifically for the former, we include 'Large Upgrade' and 'Large Downgrade', which are indicator variables for recommendation revisions in the top and bottom 20% of the sample distribution, respectively. We also include 'Top 10 Upgrade' and 'Top 10 Downgrade', which are indicator variables for upgrades or downgrades from one of the top ten brokerage firms within a month. Models 3 and 6 show that insider buying intensifies following large downgrades or downgrades initiated by analysts from top ten brokerages, suggesting that insiders' responses are more likely to be driven by signaling as those downgrades tend to have more adverse market consequences. Model 9 further indicates less insider selling following those more dramatic downgrades. Finally, although top insiders are more cautious than other insiders (as indicated by less significant estimates in Panel B), they appear to be as likely as other insiders to provide their own signals to the market in responding to analyst recommendation changes.

To summarize, the combined findings in these two sections show the dynamics between insiders and analysts are mostly one directional; analysts incorporate part of insider trading information at best and mainly for their upgrade decisions, whereas in-

Table 4

Effects of analyst recommendation revisions on subsequent insider trading. The table shows results from panel regressions of insider trading activity in the subsequent month. The dependent variable is Net Insider Trade, Insider Buy, or Insider Sell. Insider Buy (Insider Sell) is defined as the ratio of the value of insider purchases (insider sales) to the value of all insider trades. Net Insider Trade is Insider Buy minus Insider Sell. The regression results are estimated using the full sample (Panel A) and the subsample of top insiders only (Panel B) that includes CEO, President, CFO, General Counsel, and Chairman of the Board. All explanatory variables are lagged variables. The analyst recommendations are re-coded as follows: a rating of '1' reflects a strong sell recommendation, '2' a sell, '3' a hold, '4' a buy, and '5' a strong buy recommendation. Recommendation (Rec) Change is consensus analyst recommendation change from month T-2 to month T-1. Upgrade (Downgrade) is an indicator variable if Recommendation Change is positive (negative). Large Upgrade (Large Downgrade) is an indicator variables if Recommendation Change is in the top 20% (bottom 20%) of the sample distribution. Top 10 Upgrade (Top 10 Downgrade) is an indicator variable if an upgrade (a downgrade) from one of the top ten brokers is included within a month. Previous month return is the characteristics-adjusted abnormal return in month T-1. MOM is the firm's past twelve-month characteristics-adjusted abnormal return. Market value of equity (MV) is price per share times total shares outstanding and the book-to-market ratio is measured at the end of previous fiscal year. Volatility is the average daily stock return volatility in the past month. Turnover is the average daily trading volume divided by total shares outstanding in the past month. Institutional holdings is normalized by the total number of shares outstanding. Analyst Coverage is a ranking variable (1 to 5) based on the number of analysts covering the stock in the previous year. Institutional holdings are institutional ownership measured in the previous quarter. % Top 10 Brokers is the percentage of analyst recommendations from the top ten brokers for each firm in a month. Industry fixed effects (FE) are based on Fama-French 30-industry groupings, and year FEs are yearly indicator variables during the sample period except for year 2016 to ensure correct model specifications. $T-$statistics, reported in parentheses, are associated with standard errors adjusted for firm-level clustering. (***), (**), and (*) indicate significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Full Sample

	Net Insider Trade			Insider Buy			Insider Sell		
	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
Rec Change	-0.013***			-0.004***			0.009***		
	(-6.67)			(-2.93)			(6.72)		
Upgrade		-0.005***	-0.005***		-0.002**	-0.002**		0.003**	0.003**
		(-3.18)	(-2.77)		(-2.41)	(-2.10)		(2.33)	(2.02)
Downgrade		0.007***	0.003		0.002**	0.000		-0.006***	-0.003
		(5.12)	(1.45)		(2.35)	(0.26)		(-4.84)	(-1.57)
Large Upgrade			0.000			0.000			-0.000
			(0.11)			(0.16)			(-0.03)
Large Downgrade			0.007***			0.003**			-0.004**
			(2.74)			(2.08)			(-2.01)
Top 10 Upgrade			-0.001			-0.007			-0.006
			(-0.10)			(-1.37)			(-0.75)
Top 10 Downgrade			0.020**			-0.003			-0.022***
			(2.09)			(-0.52)			(-3.00)
Previous Month Return	-0.108***	-0.108***	-0.107***	-0.044***	-0.044***	-0.044***	0.063***	0.063***	0.063***
	(-32.51)	(-32.50)	(-32.45)	(-24.89)	(-24.87)	(-24.83)	(25.76)	(25.76)	(25.71)
Momentum	-0.039***	-0.039***	-0.039***	-0.007***	-0.007***	-0.007***	0.033***	0.033***	0.033***
	(-21.24)	(-21.23)	(-21.24)	(-10.40)	(-10.39)	(-10.42)	(20.60)	(20.59)	(20.59)
log(MV)	-0.021***	-0.021***	-0.021***	-0.003***	-0.003***	-0.003***	0.018***	0.018***	0.018***
	(-13.66)	(-13.67)	(-13.60)	(-4.31)	(-4.30)	(-4.22)	(12.99)	(13.00)	(12.97)
log(Book-to-Market)	0.023***	0.023***	0.023***	0.001	0.001	0.001	-0.022***	-0.022***	-0.022***
	(12.82)	(12.83)	(12.85)	(1.60)	(1.60)	(1.63)	(-13.67)	(-13.68)	(-13.68)
Volatility	0.321***	0.321***	0.320***	0.084***	0.084***	0.084***	-0.237***	-0.237***	-0.236***
	(7.64)	(7.65)	(7.62)	(3.29)	(3.30)	(3.29)	(-7.30)	(-7.29)	(-7.27)
Turnover	0.196	0.189	0.185	-0.179***	-0.179***	-0.180***	-0.375***	-0.369***	-0.366***
	(1.28)	(1.24)	(1.21)	(-2.74)	(-2.74)	(-2.76)	(-2.83)	(-2.79)	(-2.76)
Analyst Coverage	-0.006***	-0.006***	-0.006***	-0.000	-0.000	-0.000	0.006***	0.006***	0.006***
	(-4.00)	(-4.11)	(-4.03)	(-0.50)	(-0.50)	(-0.40)	(4.29)	(4.44)	(4.41)
Institutional Holdings	-0.081***	-0.081***	-0.081***	-0.018***	-0.018***	-0.018***	0.063***	0.063***	0.063***
	(-12.52)	(-12.52)	(-12.55)	(-5.68)	(-5.68)	(-5.71)	(11.11)	(11.11)	(11.13)
% Top 10 Brokers	-0.001	-0.001	-0.001	-0.003	-0.003	-0.003	-0.002	-0.002	-0.001
	(-0.17)	(-0.20)	(-0.25)	(-0.97)	(-0.98)	(-0.92)	(-0.41)	(-0.39)	(-0.29)
Intercept	0.101***	0.101***	0.100***	0.098***	0.098***	0.098***	-0.003	-0.003	-0.003
	(7.28)	(7.30)	(-7.25)	(13.66)	(13.65)	(13.56)	(-0.22)	(-0.24)	(-0.21)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	878,293	878,293	878,293	878,293	878,293	878,293	878,293	878,293	878,293
R^2	0.051	0.051	0.051	0.017	0.017	0.017	0.050	0.050	0.050

(*continued on next page*)

Table 4 (*continued*)

Panel B. Top Insiders Only

	Net Insider Trade			Insider Buy			Insider Sell		
	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
Rec Change	-0.006***			-0.002**			0.005***		
	(-5.05)			(-2.03)			(5.09)		
Upgrade		-0.002**	-0.003**		-0.001	-0.001		0.001	0.002*
		(-2.12)	(-2.37)		(-1.45)	(-1.59)		(1.64)	(1.87)
Downgrade		0.003***	-0.000		0.001	-0.000		-0.002***	-0.000
		(3.29)	(-0.06)		(1.60)	(-0.24)		(-2.92)	(-0.05)
Large Upgrade			0.002			0.000			-0.001
			(0.97)			(0.38)			(-0.93)
Large Downgrade			0.005***			0.002**			-0.003**
			(3.00)			(2.03)			(-2.36)
Top 10 Upgrade			0.001			0.000			-0.000
			(0.11)			(0.16)			(-0.04)
Top 10 Downgrade			0.011*			-0.001			-0.012**
			(1.95)			(-0.20)			(-2.47)
Previous Month Return	-0.052***	-0.052***	-0.052***	-0.020***	-0.020***	-0.020***	0.032***	0.032***	0.032***
	(-25.22)	(-25.23)	(-25.19)	(-16.88)	(-16.86)	(-16.84)	(20.16)	(20.18)	(20.15)
Momentum	-0.023***	-0.023***	-0.023***	-0.003***	-0.003***	-0.003***	0.019***	0.019***	0.019***
	(-19.57)	(-19.57)	(-19.57)	(-7.87)	(-7.86)	(-7.88)	(19.25)	(19.26)	(19.25)
log(MV)	-0.007***	-0.007***	-0.007***	-0.004***	-0.004***	-0.004***	0.003***	0.003***	0.003***
	(-7.88)	(-7.89)	(-7.78)	(-10.53)	(-10.52)	(-10.39)	(3.98)	(4.00)	(3.92)
log(Book-to-Market)	0.016***	0.016***	0.016***	0.001	0.001	0.001	-0.015***	-0.015***	-0.015***
	(13.56)	(13.56)	(13.57)	(1.62)	(1.62)	(1.64)	(-14.56)	(-14.56)	(-14.57)
Volatility	0.156***	0.156***	0.156***	0.079***	0.079***	0.079***	-0.077***	-0.077***	-0.077***
	(6.01)	(6.02)	(6.00)	(4.94)	(4.94)	(4.94)	(-3.92)	(-3.92)	(-3.90)
Turnover	-0.102	-0.104	-0.106	-0.037	-0.037	-0.038	0.065	0.067	0.069
	(-1.04)	(-1.06)	(-1.09)	(-0.93)	(-0.93)	(-0.95)	(0.76)	(0.77)	(0.79)
Analyst Coverage	-0.005***	-0.005***	-0.005***	-0.000	-0.000	-0.000	0.004***	0.005***	0.004***
	(-5.14)	(-5.20)	(-5.06)	(-0.52)	(-0.53)	(-0.42)	(5.68)	(5.76)	(5.68)
Institutional Holdings	-0.038***	-0.038***	-0.038***	-0.002	-0.002	-0.002	0.036***	0.036***	0.036***
	(-9.61)	(-9.61)	(-9.65)	(-1.43)	(-1.42)	(-1.46)	(10.17)	(10.17)	(10.20)
% Top 10 Brokers	-0.002	-0.002	-0.002	-0.000	-0.000	-0.000	0.001	0.001	0.002
	(-0.48)	(-0.49)	(-0.54)	(-0.23)	(-0.23)	(-0.22)	(0.43)	(0.44)	(0.50)
Intercept	0.040***	0.040***	0.039***	0.056***	0.057***	0.056***	0.017**	0.016**	0.017**
	(4.58)	(4.59)	(4.51)	(13.78)	(13.77)	(13.66)	(2.04)	(2.03)	(2.09)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	878,293	878,293	878,293	878,293	878,293	878,293	878,293	878,293	878,293
R^2	0.029	0.029	0.029	0.007	0.007	0.007	0.027	0.027	0.027

siders are likely to trade against analyst recommendation revisions. The former is more supportive of the biased information processing hypothesis. Although the latter is in line with both signaling and liquidity hypotheses, further evidence shows more support for the signaling effect as insider buying markedly increases following more dramatic analyst downgrades. Overall, these results do not indicate significant crowd-out effects between insiders and analysts.

5. Informativeness of insider trades and analyst recommendations

This section tests the substitution vs. complementarity hypotheses by evaluating the conditional and joint investment values of insider trades and analyst recommendations. Specifically, we employ the following specification to analyze the predictive power of insider trading and analyst recommendations on characteristic-adjusted future returns (i.e., +1M, +3M, or +6M return):

$$R_{it+m} = \alpha + \lambda_j + \tau_t + \delta(\textit{Insider Trading})_{it\ or\ it-1} + \theta(\textit{Rec Change})_{it-1\ or\ it} + \sum_k^K \beta_k X_{kit} + \epsilon_{it}, \quad (3)$$

where m=1, 3, or 6, and the variables are as defined earlier. In the above equation, we also interact insider-trading variables (Insider Buy, Insider Sell, and No Trade) with recommendation variables (Upgrade, Downgrade, and No Rec Change) to gauge the marginal effects of insiders and analysts' decisions on future stock returns when they concur or disagree with each other.[13] In addition, to explicitly consider the relative timing and informativeness of these two signals, we measure the activity of one group in month T-1 followed by that of the other group in month T, and then the returns in month T+1, T+3, and T+6. Panels A, B, and C of Table 5 present the +1M, +3M, and +6M results, respectively.

As discussed above, the premise of investigating the relative informativeness of insider trading and analyst recommendations is that each needs to be able to provide valuable information to the market alone. Therefore, we evaluate insider trading and analyst recommendations separately first (models 1–4) and then jointly (models 5–6). As shown in model 1, stocks with higher net insider trades experience significantly positive future returns. Economically, a one percentage point increase in net insider trade is associated with a 0.68% increase in the +1M abnormal return. Model 2 shows that both insider buying and selling are statistically significant at the 1% level. However, insider buying has a much more significant economic effect on the return than insider selling, indicating that the latter is less informative due to liquidity reasons. A one percentage point increase in Insider Buy is associated with a 1.18% increase in +1M return, whereas a one percentage point increase in Insider Sell is only associated with a

[13] Again, the results are based on PVB and its decomposed Insider Buy and Insider Sell ratios. The results using PNB and its decomposed ratios are reported Appendix Table A.3.

Table 5

Regressions of cumulative monthly characteristic-adjusted stock returns (in %). The table shows results from panel regressions in which the dependent variable is cumulative monthly characteristic-adjusted abnormal returns over three subsequent holding periods of one, three, and six months (1M, +3M, and +6M) following the two months in which we measure recommendation changes and then insider trading activity, or vice versa. All explanatory variables are lagged variables. Insider Buy (Insider Sell) is defined as the ratio of the value of insider purchases (insider sales) to the value of all insider trades. Net Insider Trade is Insider Buy minus Insider Sell. The analyst recommendations are re-coded as follows: a rating of '1' reflects a strong sell recommendation, '2' a sell, '3' a hold, '4' a buy, and '5' a strong buy recommendation. Rec Change is consensus analyst recommendation change either from month T-2 to T-1 or from T-1 to T, depending on whether analyst recommendation change or insider trading is measured first. Upgrade (Downgrade) is an indicator variable if Recommendation Change is positive (negative). Previous month return is the characteristics-adjusted abnormal return in month T-1. MOM is the firm's past twelve-month characteristics-adjusted abnormal return. Market value of equity (MV) is price per share times total shares outstanding and the book-to-market ratio is measured at the end of previous fiscal year. Volatility is the average daily stock return volatility in the past month. Turnover is the average daily trading volume divided by total shares outstanding in the past month. Institutional holdings is normalized by the total number of shares outstanding. Analyst Coverage is a ranking variable (1 to 5) based on the number of analysts covering the stock in the previous year. Institutional holdings are institutional ownership measured in the previous quarter. % Top 10 Brokers is the percentage of analyst recommendations from the top ten brokers for each firm in a month. In Panels B and C, control variables are included in the estimates, but their coefficients are not reported in the table. Industry fixed effects (FE) are based on Fama-French 30-industry groupings, and year FEs are yearly indicator variables during the sample period except for year 2016 to ensure correct model specifications. T−statistics, reported in parentheses, are associated with standard errors adjusted for firm-level clustering. (***), (**), and (*) indicate significance at the 1%, 5%, and 10% levels, respectively.

	Panel A. +1M					
	[1]	[2]	[3]	[4]	[5]	[6]
Net Insider Trade	0.679***					
	(20.42)					
Insider Buy		1.180***				
		(17.36)				
Insider Sell		-0.401***				
		(-10.27)				
Rec Change			0.478***			
			(5.94)			
Upgrade				-0.040		
				(-0.95)		
Downgrade				-0.184***		
				(-4.15)		
Analyst Recommendations are Measured First:						
$(\text{Upgrade})_{T-1}$*$(\text{Insider Buy})_T$					0.845***	
					(6.04)	
$(\text{Upgrade})_{T-1}$*$(\text{Insider Sell})_T$					-0.428***	
					(-5.72)	
$(\text{Downgrade})_{T-1}$*$(\text{Insider Buy})_T$					1.465***	
					(9.76)	
$(\text{Downgrade})_{T-1}$*$(\text{Insider Sell})_T$					-0.527***	
					(-7.18)	
$(\text{No Rec Change})_{T-1}$*$(\text{Insider Buy})_T$					1.189***	
					(14.07)	
$(\text{No Rec Change})_{T-1}$*$(\text{Insider Sell})_T$					-0.344***	
					(-7.08)	
Insider Trades are Measured First:						
$(\text{Insider Buy})_{T-1}$*$(\text{Upgrade})_T$						0.349**
						(2.48)
$(\text{Insider Buy})_{T-1}$*$(\text{Downgrade})_T$						0.333**
						(2.17)
$(\text{Insider Sell})_{T-1}$*$(\text{Upgrade})_T$						-0.322***
						(-4.11)
$(\text{Insider Sell})_{T-1}$*$(\text{Downgrade})_T$						-0.592***
						(-7.65)
$(\text{No Trade})_{T-1}$*$(\text{Upgrade})_T$						-0.018
						(-0.39)
$(\text{No Trade})_{T-1}$*$(\text{Downgrade})_T$						-0.146***
						(-3.00)
Previous Month Return	-0.116	-0.112	-0.191	-0.190	-0.112	-0.160
	(-0.64)	(-0.62)	(-1.06)	(-1.06)	(-0.62)	(-0.89)
Momentum	1.668***	1.662***	1.636***	1.639***	1.668***	1.645***
	(27.89)	(27.84)	(27.70)	(27.73)	(28.23)	(27.75)
log(MV)	0.361***	0.358***	0.346***	0.350***	0.356***	0.353***
	(20.36)	(20.25)	(19.75)	(19.94)	(20.19)	(20.06)
log(Book-to-Market)	0.449***	0.454***	0.462***	0.463***	0.454***	0.458***
	(16.01)	(16.21)	(16.54)	(16.57)	(16.21)	(16.39)
Volatility	8.922***	8.948***	9.224***	9.283***	8.843***	9.126***
	(4.86)	(4.87)	(5.02)	(5.06)	(4.80)	(4.97)
Turnover	-37.734***	-37.549***	-37.307***	-37.105***	-37.860***	-37.144***
	(-4.04)	(-4.02)	(-4.00)	(-3.97)	(-4.05)	(-3.97)
Analyst Coverage	-0.131***	-0.133***	-0.135***	-0.122***	-0.127***	-0.122***
	(-5.99)	(-6.07)	(-6.19)	(-5.63)	(-5.81)	(-5.63)
Institutional Holdings	-0.236***	-0.244***	-0.287***	-0.290***	-0.240***	-0.283***
	(-3.08)	(-3.19)	(-3.77)	(-3.80)	(-3.12)	(-3.71)

(*continued on next page*)

Table 5 (*continued*)

% Top 10 Brokers	-0.393***	-0.391***	-0.387***	-0.372**	-0.382***	-0.375**
	(-2.67)	(-2.65)	(-2.62)	(-2.52)	(-2.57)	(-2.54)
Intercept	-1.744***	-1.792***	-1.672***	-1.706***	-1.796***	-1.720***
	(-9.75)	(-10.03)	(-9.38)	(-9.57)	(-10.05)	(-9.64)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	876,126	876,126	876,126	876,126	876,126	876,126
R^2	0.007	0.007	0.006	0.006	0.007	0.006
Test of Marginal Effect:						
Insider Info Subsequent to Upgrade					1.273***	
					(8.09)	
Insider Info Subsequent to Downgrade					1.992***	
					(12.03)	
Insider Info Subsequent to No Rec Change					1.534***	
					(16.24)	
Analyst Info Subsequent to Insider Buy						0.017
						(0.08)
Analyst Info Subsequent to Insider Sell						0.271***
						(2.62)
Analyst Info Subsequent to No Insider Trade						0.128**
						(2.35)

	Panel B. +3M				Panel C. +6M			
	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
Insider Buy	2.333***				3.306***			
	(17.89)				(15.60)			
Insider Sell	-0.867***				-0.777***			
	(-10.51)				(-5.70)			
Upgrade		-0.206***				-0.285***		
		(-2.76)				(-2.62)		
Downgrade		-0.353***				-0.510***		
		(-4.60)				(-4.59)		
Analyst Recommendations are Measured First:								
(Upgrade)$_{T-1}$*(Insider Buy)$_{T}$			2.113***				2.417***	
			(8.27)				(6.47)	
(Upgrade)$_{T-1}$*(Insider Sell)$_{T}$			-0.976***				-0.969***	
			(-7.21)				(-4.74)	
(Downgrade)$_{T-1}$*(Insider Buy)$_{T}$			2.216***				3.519***	
			(8.71)				(9.05)	
(Downgrade)$_{T-1}$*(Insider Sell)$_{T}$			-0.970***				-1.083***	
			(-7.12)				(-5.21)	
(No Rec Change)$_{T-1}$*(Insider Buy)$_{T}$			2.429***				3.472***	
			(15.32)				(13.75)	
(No Rec Change)$_{T-1}$*(Insider Sell)$_{T}$			-0.790***				-0.593***	
			(-7.90)				(-3.71)	
Insider Trades are Measured First:								
(Insider Buy)$_{T-1}$*(Upgrade)$_{T}$				0.954***				1.900***
				(3.82)				(5.19)
(Insider Buy)$_{T-1}$*(Downgrade)$_{T}$				0.863***				1.302***
				(3.28)				(3.35)
(Insider Sell)$_{T-1}$*(Upgrade)$_{T}$				-0.753***				-0.850***
				(-5.39)				(-4.10)
(Insider Sell)$_{T-1}$*(Downgrade)$_{T}$				-1.033***				-0.980***
				(-7.46)				(-4.74)
(No Trade)$_{T-1}$*(Upgrade)$_{T}$				-0.184**				-0.352***
				(-2.35)				(-2.89)
(No Trade)$_{T-1}$*(Downgrade)$_{T}$				-0.318***				-0.567***
				(-3.72)				(-4.55)
Intercept	-4.866***	-4.714***	-4.880***	-4.735***	-7.129***	-6.916***	-7.195***	-6.926***
	(-9.55)	(-9.25)	(-9.60)	(-9.29)	(-7.33)	(-7.10)	(-7.42)	(-7.11)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	876,126	876,126	876,126	876,126	876,126	876,126	876,126	876,126
R^2	0.012	0.011	0.012	0.012	0.011	0.010	0.011	0.010
Test of Marginal Effect:								
Insider Info Subsequent to Upgrade			3.089***				3.386***	
			(10.76)				(8.00)	
Insider Info Subsequent to Downgrade			3.185***				4.601***	
			(10.94)				(10.44)	
Insider Info Subsequent to No Rec Change			3.220***				4.066***	
			(17.67)				(14.12)	
Analyst Info Subsequent to Insider Buy				0.091				0.598
				(0.27)				(1.23)
Analyst Info Subsequent to Insider Sell				0.279				0.130
				(1.62)				(0.55)
Analyst Info Subsequent to No Insider Trade				0.124				0.214
				(1.27)				(1.57)

Table 6

Regressions of cumulative monthly characteristic-adjusted stock returns (in %): Top insiders only. The table shows results from panel regressions using the subsample of top insiders only. Top insiders include CEO, President, CFO, General Counsel, and Chairman of the Board. The dependent variable is cumulative monthly characteristic-adjusted abnormal returns over three subsequent holding periods of one, three, and six months (1M, +3M, and +6M) following the two months in which we measure recommendation changes and then insider trading activity, or vice versa. All explanatory variables are lagged variables. Insider Buy (Insider Sell) is defined as the ratio of the value of insider purchases (insider sales) to the value of all insider trades. Net Insider Trade is Insider Buy minus Insider Sell. The analyst recommendations are re-coded as follows: a rating of ‘1’ reflects a strong sell recommendation, ‘2’ a sell, ‘3’ a hold, ‘4’ a buy, and ‘5’ a strong buy recommendation. Rec Change is consensus analyst recommendation change either from month T-2 to T-1 or from T-1 to T, depending on whether analyst recommendation change or insider trading is measured first. Upgrade (Downgrade) is an indicator variable if Recommendation Change is positive (negative). Previous month return is the characteristics-adjusted abnormal return in month T-1. MOM is the firm’s past twelve-month characteristics-adjusted abnormal return. Market value of equity (MV) is price per share times total shares outstanding and the book-to-market ratio is measured at the end of previous fiscal year. Volatility is the average daily stock return volatility in the past month. Turnover is the average daily trading volume divided by total shares outstanding in the past month. Institutional holdings is normalized by the total number of shares outstanding. Analyst Coverage is a ranking variable (1 to 5) based on the number of analysts covering the stock in the previous year. Institutional holdings are institutional ownership measured in the previous quarter. % Top 10 Brokers is the percentage of analyst recommendations from the top ten brokers for each firm in a month. Control variables are included in the estimates, but their coefficients are not reported in the table. Industry fixed effects (FE) are based on Fama-French 30-industry groupings, and year FEs are yearly indicator variables during the sample period except for year 2016 to ensure correct model specifications. $T-$statistics, reported in parentheses, are associated with standard errors adjusted for firm-level clustering. (***), (**), and (*) indicate significance at the 1%, 5%, and 10% levels, respectively.

	+1M			+3M			+6M		
	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
Insider Buy	1.690***			3.033***			4.182***		
	(15.10)			(14.61)			(15.60)		
Insider Sell	-0.482***			-1.135***			-1.173***		
	(-8.28)			(-9.25)			(-5.70)		
Analyst Recommendations are Measured First:									
$(\text{Upgrade})_{T-1}$*$(\text{Insider Buy})_T$		1.428***			3.224***			3.494***	
		(5.47)			(6.74)			(5.01)	
$(\text{Upgrade})_{T-1}$*$(\text{Insider Sell})_T$		-0.438***			-1.199***			-1.278***	
		(-3.56)			(-5.58)			(-3.93)	
$(\text{Downgrade})_{T-1}$*$(\text{Insider Buy})_T$		2.359***			4.458***			6.264***	
		(8.10)			(9.28)			(8.68)	
$(\text{Downgrade})_{T-1}$*$(\text{Insider Sell})_T$		-0.575***			-1.102***			-1.346***	
		(-5.06)			(-5.19)			(-4.11)	
$(\text{No Rec Change})_{T-1}$*$(\text{Insider Buy})_T$		1.584***			2.664***			3.831***	
		(11.73)			(10.86)			(9.87)	
$(\text{No Rec Change})_{T-1}$*$(\text{Insider Sell})_T$		-0.462***			-1.120***			-1.058***	
		(-6.21)			(-7.44)			(-4.38)	
Insider Trades are Measured First:									
$(\text{Insider Buy})_{T-1}$*$(\text{Upgrade})_T$			0.618**			1.574***			2.451***
			(2.31)			(3.47)			(3.61)
$(\text{Insider Buy})_{T-1}$*$(\text{Downgrade})_T$			0.415			1.999***			2.891***
			(1.49)			(4.21)			(4.16)
$(\text{Insider Sell})_{T-1}$*$(\text{Upgrade})_T$			-0.335***			-0.865***			-1.229***
			(-2.75)			(-3.95)			(-3.80)
$(\text{Insider Sell})_{T-1}$*$(\text{Downgrade})_T$			-0.599***			-1.024***			-1.270***
			(-4.99)			(-4.66)			(-3.95)
$(\text{No Trade})_{T-1}$*$(\text{Upgrade})_T$			0.012			-0.138*			-0.328***
			(0.26)			(-1.70)			(-2.74)
$(\text{No Trade})_{T-1}$*$(\text{Downgrade})_T$			-0.117**			-0.262***			-0.543***
			(-2.46)			(-3.13)			(-4.41)
Intercept	-1.762***	-1.750***	-1.693***	-4.787***	-4.752***	-4.688***	-7.019***	-7.012***	-6.906***
	(-9.85)	(-9.79)	(-9.49)	(-9.39)	(-9.35)	(-9.20)	(-7.21)	(-7.23)	(-7.10)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	876,126	876,126	876,126	876,126	876,126	876,126	876,126	876,126	876,126
R^2	0.006	0.007	0.006	0.012	0.012	0.012	0.011	0.011	0.010
Test of Marginal Effect:									
Insider Info Subsequent to Upgrade		1.866***			4.423***			4.772***	
		(6.51)			(8.52)			(6.19)	
Insider Info Subsequent to Downgrade		2.934***			5.560***			7.610***	
		(9.42)			(10.54)			(9.60)	
Insider Info Subsequent to No Rec Change		2.046***			3.784***			4.889***	
		(13.52)			(13.24)			(10.83)	
Analyst Info Subsequent to Insider Buy			0.203			-0.425			-0.440
			(0.52)			(-0.66)			(-0.48)
Analyst Info Subsequent to Insider Sell			0.264			0.159			0.042
			(1.59)			(0.55)			(0.11)
Analyst Info Subsequent to No Insider Trade			0.128**			0.124			0.215
			(2.35)			(1.27)			(1.57)

0.40% decrease in returns. Model 3 shows that analyst recommendation revisions also provide predictive power for future returns; a stock with an increase in its consensus recommendation rating by one notch is positively associated with 0.48% abnormal return. This result is mainly driven by downgrades but not upgrades, as revealed in model 4. Downgrades are associated with a 0.18% decline in return, while upgrades have statistically insignificant future investment value. Note that the coefficient estimates for Upgrade and Downgrade are based on consensus recommendation revisions. Thus, these two indicator variables, to some extent, estimate the average of the marginal effects of all upgrades and downgrades within a month on future returns, and they could vary significantly, depending on the level of analyst coverage and individual analysts.[14]

Provided that both insider trading (buying and selling) and analyst recommendations (downgrades) are informative when evaluated separately, we next examine the conditional informativeness of these two signals. Note that since analyst upgrades are shown uninformative, our subsequent discussions focus on analyst downgrades and their interactions with insider trading therein. In models 5 and 6, analyst recommendation revisions are measured prior to and following insider trading, respectively. We further report the marginal effect of one signal conditional on the other at the bottom of the panel. Several critical patterns emerge. First, the estimates in model 5 indicate that insider trading continues to provide significant information beyond that of analyst recommendations. For example, the marginal effect of insider trading conditional on prior-month analyst downgrades is 1.99%, significant at the 1% level, suggesting that insider trading provides valuable information to the market beyond the information by analyst recommendations. Further, analyst recommendations also provide significant investment value beyond that of insider trading. As shown in model 6, the marginal effect of analyst recommendations conditional on prior-month insider selling is 0.27%.[15]

Second, we further distinguish between the complementarity and substitution effects by comparing the joint versus individual informativeness of these two signals. Our estimates show that the market reaction to the combined concurring negative signals (analyst downgrades and insider selling) is lower than the reaction to each of the individual signals alone: -0.53% and -0.59% (combined in models 5 and 6, respectively) vs. -0.18% (downgrades) and -0.40% (insider sell). On the other hand, when these two signals disagree, their joint effect is fairly mitigated. For instance, the joint return effect of insider buying and analyst downgrades in model 6 is 0.33%, which falls between the individual return effects of downgrades (-0.18%) and insider buying (1.18%). These additional findings provide further support for the complementary effect.

Third, although the overall evidence is more in line with the complementarity effect, our estimates do unveil the dominant signal of insider buying, which could effectively substitute the information content of analyst recommendations. For example, model 6 shows that the marginal effect of analyst recommendations conditional on prior-month insider buying is insignificant as future returns tend to be positive following insiders' purchases of their firm's stocks irrespective of analyst recommendations (0.35% for upgrades vs. 0.33% for downgrades). In addition, model 5 indicates that analyst downgrades combined with subsequent insider buying is associated with a significant 1.47% return, which is even higher than the return associated with insider buying alone (1.18% in model 2). This result implies that insider buying following analyst downgrades are associated with strong signaling effects and could stymie the negative information from analyst downgrades.[16]

We repeat models 2, 4, 5, and 6 in Panel A for +3M (Panel B) and +6M (Panel C) returns. Again, when evaluated separately (models 1, 2, 5, and 6), both insider trading (buying and selling) and analyst downgrades continue to be informative. Interestingly, analyst upgrades are associated with significant return reversals, implying that either investors are overly optimistic about upgrades when initially announced or the informativeness associated with upgrades is mostly short term. The remaining models evaluate these two groups of informed agents jointly or conditionally on each other's signals. Consistent with the complementarity effect, insider trading continues to provide valuable information to the market for a longer term following analyst recommendation revisions. Further, regardless of which signal is measured first, the joint concurring effect of insider selling and analyst downgrades remains more severe than each of these two negative signals alone. However, the positive signaling effect of insider buying continues to dominate in the next six months, especially following analyst downgrades, a finding more in line with the substitution hypothesis.

Next, we examine whether the documented informativeness of insider trading and analyst recommendations differs for top insiders. Several interesting findings emerge from Table 6. First, when evaluated alone, the trading activity of top insiders produces stronger signals than that of all insiders. For example, a one percentage point increase in top insider buying leads to a 1.69% increase in +1M return, compared to 1.18% for all insiders. The difference is even more dramatic for +3M and +6M returns: 3.03% and 4.18% for top insiders versus 2.33% and 3.31% for all insiders. Top insider selling is also associated with more negative future returns than all insider selling, though top insider selling continues to have much smaller return effects than buying. As such, we find that conditional on analyst recommendations, the information content of top insider trading is more economically significant, substantiated by the estimates of the marginal effect. In addition, although the joint concurring effect of insider selling and analyst downgrades remains more negative than each of the two signals alone in most cases, the joint effect is closer to the single negative effect of insider selling, implying that the effect from analyst downgrades is less significant when we include only top insiders than when we include all insiders in the sample. Nonetheless, the combined evidence from top insiders continues to lend support for the complementarity hypothesis.

Finally, the singular substitution effect of top insider buying following analyst downgrades is also more evident than that of all insider buying. Models 2, 5, and 8 reveal large abnormal returns of 2.36% (+1M), 4.46% (+3M), and 6.26% (+6M) associated with those cases. This evidence suggests that the signaling effect of top insider buying is more credible than that of other insiders. Overall, this section shows differing informativeness of insider trading and analyst recommendations. Consistent with the complementarity hypothesis, both insider trading and analyst recommendations could provide value-relevant information to the markets. However, the strong signaling effect of insider buying following analyst downgrades could substitute the informativeness of those downgrades.

[14] We also include in the regressions four additional analyst-related variables as described above. The results, reported in Panel A of Appendix Table A.4, show that although the return effect of upgrades in general is uninformative or even reversed, large upgrades are significantly informative.

[15] The results in Panel C of Appendix Table A.4 indicate that this complementarity effect could be driven by large analyst recommendation revisions, especially downgrades, following insider selling.

[16] Panel B of Appendix Table A.4 further shows that insider buying is significantly more informative than insider selling subsequent to analysts' large downgrades, again supporting insiders' strong signaling effects following those large downgrades (1.70% for insider buying vs. -0.17% for insider selling.)

6. Additional robustness tests

6.1. Post 2002

There were three important legislations after 2000 related to insider trading and analyst recommendations: RegFD in 2000, SOX (Sarbanes-Oxley Act) in 2002, and GARS in 2002. SOX requires insiders to report their trades within two business days rather than ten days after the end of the trading month. Following SOX, insider trading has become more informative as the information is disseminated more timely (e.g., Brochet, 2010). SOX, however, could also make insiders more reluctant to trade if their trades are deemed taking advantage of their informational advantage at the expense of other shareholders. For RegFD and GARS, they are enacted to, respectively, curtail selective disclosures by corporate executives and minimize conflicts of interest between analyst research and investment banking business. Enhanced regulation could either stifle analysts' ability to provide valuable recommendations due to limited access to private information sources or increase the informativeness of analyst recommendations due to reduced conflicts of interest. Existing studies indicate that post RegFD and GARS, the information content of analyst research is mostly reduced.[17]

In our context, we utilize these three regulations as exogenous shocks to investigate whether they significantly reshape the documented relations between insider trades and analyst recommendations. Specifically, we repeat our analyses with a restricted sample period of 2003–2016 and report the results in Table 7. Consistent with the full sample results, Panel A shows mixed patterns post-2002 between insider trading and subsequent analyst recommendation revisions. For instance, while there is some weak evidence that insider buying is associated with analyst upgrade (model 2), we also find that both insider buying and selling are associated with a lower likelihood of analyst upgrades and downgrades (models 4 and 6). Consequently, the post-2002 results continue to suggest that analysts do not fully consider insider trading when revising their recommendations. Further, Panel B continues to show the contrasting pattern of subsequent insider trading vis-*à*-vis analyst recommendations post-2002, as indicated in the first three models. Insiders are also less likely to buy following upgrades or sell following downgrades (models 5 and 8). Interestingly, the remaining models indicate that post-2002, insiders are less likely to trade following large analyst recommendation revisions or revisions by top ten analysts, implying increased litigation risk could significantly affect insider trading decisions. Overall, the results in this panel suggest that insiders' motives are likely driven by signaling or liquidity rather than information.

Finally, Panel C reveals that when evaluated individually, insider trading (buying and selling) and analyst downgrades continue to provide value-relevant information to the markets even though the effects from both signals have reduced post-2002. Such lowered impacts implies the effectiveness of regulations on insider trading and brokerage industry. Additionally, insider buying has a larger economic impact than selling, implying liquidity reasons drive a large portion of the latter. However, the effect of analyst downgrades has become more long-term in recent years as only the +3M and +6M returns are significant. Moreover, conditional on analyst recommendations, subsequent insider trading continues to exhibit economic importance post-2002. The joint concurring effect of insider selling and analyst downgrades generally is also more negative than that of each of the two signals alone. Thus, the post-2002 findings remain consistent with the complementarity hypothesis. However, the post-2002 results continue to support a singular substitution effect of insider buying on analyst recommendations; following analyst downgrades, insider buying is associated with significantly positive returns of 0.52% (+1M), 0.77% (+3M), and 2.27% (+6M).

6.2. Excluding earnings announcement (EA) months

There is some concern that our results may simply reflect insiders' and analysts' reactions to EAs. For example, Altinkilic and Hansen (2009) and Yezegel (2015) document that analysts often revise their recommendations in response to significant firm-specific news, especially EAs. In these situations, insiders' and analysts' actions may contain limited valuable information for other investors. However, it is unlikely that our evidence is driven by insiders' and analysts' responses to EAs for two reasons. First, insiders are refrained from trading on firm-specific information within a certain window around the news release dates because of regulations, heightened litigation risk, and explicit corporate policies. Second, analyst recommendation revisions remain informative following EAs (e.g., Bettis et al., 2000; Li et al., 2015; Yezegel, 2015). Nevertheless, to rule out this concern, we re-run our main tests with a subsample that excludes EA months and report the regression estimates in Table 8. Overall, the results are qualitatively similar to our main findings. Panel A indicates that although there is some evidence that analysts take into account insider buying when revising their recommendations, their revision decisions as a whole are not associated with the directions of insider trading in a systematic way. Panel B shows that outside the EA months, insiders also trade in the opposite directions of analyst recommendation revisions due to signaling reasons. Finally, Panel C shows that our documented results of the complementarity effect of insider trading and analyst downgrades as well as the singular substitution effect of insider buying remain materially unaffected outside EA months.

6.3. Before and after EA months

In this section, we conduct another extension to shed more light on the relative informativeness of insider trading and analyst recommendations. Specifically, we again exploit EAs during which the informativeness of one signal is markedly shocked. However, instead of utilizing the subsample of non-EA months, we re-run our tests with two separate subsamples that only include observations in the month prior to and following the EA month, respectively. This EA setting is ideal for our analysis since the information environment is greatly altered around an EA. Further, though it is not one of the objectives of this paper, it would be interesting to examine whether the return predictability of analyst recommendations stems from analysts' ability to either exploit private information before EAs or skillfully analyze public information following EAs, and whether such ability is affected by insider trading.[18]

Table 9 reports the main findings. First, when evaluated separately, insider trading is informative both before and after EAs although the former is more economically significant than the latter. Analyst downgrades are also informative in both periods except the +1M before the EA month, and in contrast to insider trading, downgrades seem more economically significant after than

[17] For example, Gintschel and Markov (2004) find that post RegFD, the absolute price impact of analyst forecasts, and recommendations are considerably lower. Agrawal et al. (2006) also document that analyst forecasts become less accurate, and forecast dispersion increases post RegFD. Further, Kadan et al. (2009) find that following GARS, analyst recommendations become less informative though optimistic recommendations instead become more informative.

[18] In a similar vein, there is an ongoing debate regarding whether another group of information providers, short sellers, obtain their ability through public versus private information. See, e.g., Francis et al. (2005); Christophe et al. (2010); Karpoff and Lou (2010); Engelberg et al. (2012); Boehmer et al. (2020); Christophe and Hsieh (2023).

Table 7

Insider trades and analyst recommendation revisions: Post-2002. The table summarizes additional regression results on the relationship between insider trading activity and analyst recommendation revisions using a subsample that restricts the sample period to post SOX, RegFD, and GARS, i.e., from 2003 to 2016. Insider Buy (Insider Sell) is defined as the ratio of the value of insider purchases (insider sales) to the value of all insider trades. Net Insider Trade is Insider Buy minus Insider Sell. The analyst recommendations are re-coded as follows: a rating of ‘1’ reflects a strong sell recommendation, ‘2’ a sell, ‘3’ a hold, ‘4’ a buy, and ‘5’ a strong buy recommendation. Rec Change is consensus analyst recommendation change either from month T-2 to T-1 or from T-1 to T, depending on whether analyst recommendation change or insider trading is measured first. Upgrade (Downgrade) is an indicator variable if Recommendation Change is positive (negative). Large Upgrade (Large Downgrade) is an indicator variables if Recommendation Change is in the top 20% (bottom 20%) of the sample distribution. Top 10 Upgrade (Top 10 Downgrade) is an indicator variable if an upgrade (a downgrade) from one of the top ten brokers is included within a month. The dependent variable in Panel C is cumulative monthly characteristic-adjusted abnormal returns over three subsequent holding periods of one, three, and six months (1M, +3M, and +6M) following the two months in which we measure recommendation changes and then insider trading activity, or vice versa. All explanatory variables are lagged variables. Previous month return is the characteristics-adjusted abnormal return in month T-1. Momentum is the firm’s past twelve-month characteristics-adjusted abnormal return. Market value of equity (MV) is price per share times total shares outstanding and the book-to-market ratio is measured at the end of previous fiscal year. Volatility is the average daily stock return volatility in the past month. Turnover is the average daily trading volume divided by total shares outstanding in the past month. Institutional holdings is normalized by the total number of shares outstanding. Analyst Coverage is a ranking variable (1 to 5) based on the number of analysts covering the stock in the previous year. Institutional holdings are institutional ownership measured in the previous quarter. % Top 10 Brokers is the percentage of analyst recommendations from the top ten brokers for each firm in a month. Control variables are included in the estimates, but their coefficients are not reported in the table. Industry fixed effects (FE) are based on Fama-French 30-industry groupings, and year FEs are yearly indicator variables during the sample period except for year 2016 to ensure correct model specifications. $T-$statistics, reported in parentheses, are associated with standard errors adjusted for firm-level clustering. (***), (**), and (*) indicate significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Effect of Insider Trading on Subsequent Analyst Recommendation Revisions

	Rec Change		Upgrade		Downgrade	
	[1]	[2]	[3]	[4]	[5]	[6]
Net Insider Trade	0.001**		0.013		0.011	
	(2.25)		(1.54)		(1.28)	
Insider Buy		0.005***		-0.044**		-0.124***
		(4.41)		(-2.49)		(-6.97)
Insider Sell		0.001		-0.034***		-0.060***
		(0.80)		(-3.37)		(-5.93)
Intercept	-0.002	-0.002	-3.725***	-3.722***	-3.673***	-3.667***
	(-0.97)	(-1.09)	(-76.67)	(-76.25)	(-86.79)	(-86.05)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	489,076	489,076	489,076	489,076	489,076	489,076
R^2	0.002	0.002				

Panel B. Effect of Analyst Recommendation Revisions on Subsequent Insider Trading

	Net Insider Trade			Insider Buy			Insider Sell		
	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
Rec Change	-0.011***			-0.001			0.009***		
	(-3.98)			(-0.79)			(4.62)		
Upgrade		-0.005**	-0.005**		-0.002**	-0.001		0.003	0.004**
		(-2.31)	(-2.00)		(-2.22)	(-0.61)		(1.49)	(1.99)
Downgrade		0.006***	0.005*		-0.000	0.002		-0.006***	-0.003
		(3.11)	(1.80)		(-0.03)	(1.37)		(-3.77)	(-1.37)
Large Upgrade			0.001			-0.004**			-0.005
			(0.16)			(-2.29)			(-1.54)
Large Downgrade			0.002			-0.003*			-0.005
			(0.46)			(-1.78)			(-1.57)
Top 10 Upgrade			0.004			0.002			-0.002
			(0.24)			(0.22)			(-0.16)
Top 10 Downgrade			0.023			-0.007			-0.029**
			(1.53)			(-0.92)			(-2.36)
Intercept	0.143***	0.143***	0.143***	0.096***	0.096***	0.097***	-0.047**	-0.048**	-0.046**
	(6.54)	(6.54)	(6.54)	(12.33)	(12.30)	(12.36)	(-2.38)	(-2.40)	(-2.34)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	488,643	488,643	488,643	488,643	488,643	488,643	488,643	488,643	488,643
R^2	0.055	0.055	0.055	0.021	0.021	0.021	0.052	0.052	0.052

(*continued on next page*)

Table 7 (*continued*)

Panel C. Regressions of Cumulative Monthly Characteristic-Adjusted Returns (in %)

	+1M				+3M				+6M			
	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	[11]	[12]
Insider Buy	0.541***				1.399***				2.305***			
	(6.90)				(9.04)				(8.83)			
Insider Sell	-0.295***				-0.678***				-0.543***			
	(-7.13)				(-7.36)				(-3.48)			
Upgrade		-0.049				-0.148*				-0.399***		
		(-0.98)				(-1.67)				(-3.07)		
Downgrade		-0.057				-0.172*				-0.396***		
		(-1.09)				(-1.92)				(-3.10)		
<u>*Analyst Recommendations are Measured First:*</u>												
(Upgrade)$_{T-1}$*(Insider Buy)$_{T}$			0.457**				1.626***				1.587***	
			(2.50)				(4.77)				(3.15)	
(Upgrade)$_{T-1}$*(Insider Sell)$_{T}$			-0.345***				-0.936***				-0.878***	
			(-4.46)				(-6.36)				(-3.84)	
(Downgrade)$_{T-1}$*(Insider Buy)$_{T}$			0.517***				0.768**				2.267***	
			(2.79)				(2.27)				(4.23)	
(Downgrade)$_{T-1}$*(Insider Sell)$_{T}$			-0.333***				-0.685***				-0.735***	
			(-4.48)				(-4.53)				(-3.07)	
(No Rec Change)$_{T-1}$*(Insider Buy)$_{T}$			0.571***				1.516***				2.485***	
			(6.04)				(8.08)				(8.20)	
(No Rec Change)$_{T-1}$*(Insider Sell)$_{T}$			-0.261***				-0.580***				-0.349**	
			(-5.01)				(-5.30)				(-1.99)	
<u>*Insider Trades are Measured First:*</u>												
(Insider Buy)$_{T-1}$*(Upgrade)$_{T}$				0.061				0.478				1.204**
				(0.31)				(1.37)				(2.43)
(Insider Buy)$_{T-1}$*(Downgrade)$_{T}$				0.122				0.348				0.299
				(0.61)				(1.00)				(0.56)
(Insider Sell)$_{T-1}$*(Upgrade)$_{T}$				-0.197**				-0.436***				-0.573**
				(-2.38)				(-2.90)				(-2.54)
(Insider Sell)$_{T-1}$*(Downgrade)$_{T}$				-0.324***				-0.538***				-0.646***
				(-3.93)				(-3.52)				(-2.81)
(No Trade)$_{T-1}$*(Upgrade)$_{T}$				-0.009				-0.087				-0.411***
				(-0.16)				(-0.88)				(-2.77)
(No Trade)$_{T-1}$*(Downgrade)$_{T}$				-0.002				-0.109				-0.363**
				(-0.03)				(-1.09)				(-2.49)
Intercept	-1.328***	-1.274***	-1.348***	-1.096***	-4.192***	-4.060***	-4.253***	-3.586***	-6.130***	-5.969***	-6.287***	-5.308***
	(-6.56)	(-6.36)	(-6.68)	(-7.40)	(-6.92)	(-6.73)	(-7.09)	(-9.22)	(-5.15)	(-5.02)	(-5.35)	(-8.07)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	485,469	485,469	485,469	485,469	485,469	485,469	485,469	485,469	485,469	485,469	485,469	485,469
R^2	0.006	0.005	0.006	0.005	0.011	0.011	0.011	0.010	0.010	0.010	0.010	0.009
<u>*Test of Marginal Effect:*</u>												
Insider Info Subsequent to Upgrade			0.802***				2.562***				2.465***	
			(4.07)				(6.90)				(4.44)	
Insider Info Subsequent to Downgrade			0.850***				1.454***				3.002***	
			(4.29)				(3.92)				(5.12)	
Insider Info Subsequent to No Rec Change			0.833***				2.095***				2.834***	
			(8.03)				(9.92)				(8.35)	
Analyst Info Subsequent to Insider Buy				-0.060				0.130				0.905
				(-0.21)				(0.28)				(1.36)
Analyst Info Subsequent to Insider Sell				0.127				0.102				0.073
				(1.16)				(0.55)				(0.29)
Analyst Info Subsequent to No Insider Trade				-0.007				0.021				-0.048
				(-0.11)				(0.18)				(-0.28)

Table 8

Insider trades and analyst recommendation revisions using the subsample that excludes earnings announcement months. The table summarizes additional regression results on the relationship between insider trading activity and analyst recommendation revisions using a subsample that excludes the months of earnings announcements. Insider Buy (Insider Sell) is defined as the ratio of the value of insider purchases (insider sales) to the value of all insider trades. Net Insider Trade is Insider Buy minus Insider Sell. The analyst recommendations are re-coded as follows: a rating of ‘1’ reflects a strong sell recommendation, ‘2’ a sell, ‘3’ a hold, ‘4’ a buy, and ‘5’ a strong buy recommendation. Rec Change is consensus analyst recommendation change either from month T-2 to T-1 or from T-1 to T, depending on whether analyst recommendation change or insider trading is measured first. Upgrade (Downgrade) is an indicator variable if Recommendation Change is positive (negative). Large Upgrade (Large Downgrade) is an indicator variables if Recommendation Change is in the top 20% (bottom 20%) of the sample distribution. Top 10 Upgrade (Top 10 Downgrade) is an indicator variable if an upgrade (a downgrade) from one of the top ten brokers is included within a month. The dependent variable in Panel C is cumulative monthly characteristic-adjusted abnormal returns over three subsequent holding periods of one, three, and six months (1M, +3M, and +6M) following the two months in which we measure recommendation changes and then insider trading activity, or vice versa. All explanatory variables are lagged variables. Previous month return is the characteristics-adjusted abnormal return in month T-1. Momentum is the firm’s past twelve-month characteristics-adjusted abnormal return. Market value of equity (MV) is price per share times total shares outstanding and the book-to-market ratio is measured at the end of previous fiscal year. Volatility is the average daily stock return volatility in the past month. Turnover is the average daily trading volume divided by total shares outstanding in the past month. Institutional holdings is normalized by the total number of shares outstanding. Analyst Coverage is a ranking variable (1 to 5) based on the number of analysts covering the stock in the previous year. Institutional holdings are institutional ownership measured in the previous quarter. % Top 10 Brokers is the percentage of analyst recommendations from the top ten brokers for each firm in a month. Control variables are included in the estimates, but their coefficients are not reported in the table. Industry fixed effects (FE) are based on Fama-French 30-industry groupings, and year FEs are yearly indicator variables during the sample period except for year 2016 to ensure correct model specifications. $T-$statistics, reported in parentheses, are associated with standard errors adjusted for firm-level clustering. (***), (**), and (*) indicate significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Effect of Insider Trading on Subsequent Analyst Recommendation Revisions

	Rec Change		Upgrade		Downgrade	
	[1]	[2]	[3]	[4]	[5]	[6]
Net Insider Trade	0.001		0.022***		0.006	
	(1.38)		(2.96)		(0.86)	
Insider Buy		0.003**		0.002		-0.038***
		(2.46)		(0.10)		(-2.79)
Insider Sell		0.000		-0.033***		-0.029***
		(0.49)		(-3.44)		(-3.03)
Intercept	0.002	0.002	-4.259***	-4.257***	-4.048***	-4.044***
	(0.97)	(0.90)	(-92.20)	(-92.06)	(-93.92)	(-93.60)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	614,624	614,624	614,624	614,624	614,624	614,624
R^2	0.003	0.003				

Panel B. Effect of Analyst Recommendation Revisions on Subsequent Insider Trading

	Net Insider Trade			Insider Buy			Insider Sell		
	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
Rec Change	-0.013***			-0.003**			0.010***		
	(-6.06)			(-2.46)			(6.20)		
Upgrade		-0.004**	-0.005**		-0.002*	-0.002*		0.003*	0.003
		(-2.44)	(-2.12)		(-1.94)	(-1.86)		(1.74)	(1.41)
Downgrade		0.008***	0.003		0.002*	-0.001		-0.006***	-0.003
		(4.77)	(1.06)		(1.74)	(-0.46)		(-4.77)	(-1.52)
Large Upgrade			-0.000			0.001			0.001
			(-0.04)			(0.45)			(0.35)
Large Downgrade			0.009***			0.004**			-0.005**
			(3.05)			(2.43)			(-2.17)
Top 10 Upgrade			-0.003			-0.010*			-0.007
			(-0.28)			(-1.86)			(-0.80)
Top 10 Downgrade			0.010			-0.002			-0.012
			(0.93)			(-0.35)			(-1.38)
Intercept	0.096***	0.096***	0.095***	0.083***	0.083***	0.082***	-0.013	-0.013	-0.013
	(6.89)	(6.91)	(6.84)	(11.54)	(11.52)	(11.40)	(-0.98)	(-1.02)	(-0.98)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	612,472	612,472	612,472	612,472	612,472	612,472	612,472	612,472	612,472
R^2	0.050	0.050	0.050	0.018	0.018	0.018	0.049	0.049	0.049

(*continued on next page*)

Table 8 (*continued*)

Panel C. Regressions of Cumulative Monthly Characteristic-Adjusted Returns (in %)

	+1M				+3M				+6M			
	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	[11]	[12]
Insider Buy	1.221***				2.309***				3.273***			
	(14.09)				(14.70)				(13.00)			
Insider Sell	-0.406***				-0.851***				-0.752***			
	(-8.22)				(-9.16)				(-5.00)			
Upgrade		-0.044				-0.196**				-0.148		
		(-0.82)				(-2.18)				(-1.12)		
Downgrade		-0.206***				-0.359***				-0.466***		
		(-3.66)				(-3.87)				(-3.49)		
Analyst Recommendations are Measured First:												
$(\text{Upgrade})_{T-1}$*$(\text{Insider Buy})_T$			0.951***				2.116***				2.473***	
			(5.29)				(6.89)				(5.51)	
$(\text{Upgrade})_{T-1}$*$(\text{Insider Sell})_T$			-0.449***				-0.957***				-0.969***	
			(-4.85)				(-6.04)				(-4.08)	
$(\text{Downgrade})_{T-1}$*$(\text{Insider Buy})_T$			1.515***				2.045***				3.300***	
			(7.78)				(6.67)				(7.14)	
$(\text{Downgrade})_{T-1}$*$(\text{Insider Sell})_T$			-0.550***				-0.997***				-1.000***	
			(-6.07)				(-6.37)				(-4.27)	
$(\text{No Rec Change})_{T-1}$*$(\text{Insider Buy})_T$			1.209***				2.469***				6.495***	
			(11.19)				(12.61)				(11.34)	
$(\text{No Rec Change})_{T-1}$*$(\text{Insider Sell})_T$			-0.333***				-0.776***				-0.595***	
			(-5.28)				(-6.59)				(-3.21)	
Insider Trades are Measured First:												
$(\text{Insider Buy})_{T-1}$*$(\text{Upgrade})_T$				0.394***				1.009***				2.024***
				(2.33)				(3.53)				(4.81)
$(\text{Insider Buy})_{T-1}$*$(\text{Downgrade})_T$				0.413**				1.194***				1.849***
				(2.32)				(3.99)				(4.28)
$(\text{Insider Sell})_{T-1}$*$(\text{Upgrade})_T$				-0.357***				-0.832***				-0.758***
				(-3.81)				(-5.23)				(-3.28)
$(\text{Insider Sell})_{T-1}$*$(\text{Downgrade})_T$				-0.616***				-1.150***				-0.925***
				(-6.61)				(-7.22)				(-4.00)
$(\text{No Trade})_{T-1}$*$(\text{Upgrade})_T$				-0.016				-0.159				-0.193
				(-0.27)				(-1.56)				(-1.28)
$(\text{No Trade})_{T-1}$*$(\text{Downgrade})_T$				-0.173***				-0.318***				-0.559***
				(-2.75)				(-2.98)				(-3.64)
Intercept	-1.985***	-1.914***	-1.997***	-1.930***	-5.320***	-5.120***	-4.899***	-4.783***	-7.472***	-7.192***	-7.216***	-6.967***
	(-8.92)	(-8.62)	(-8.93)	(-8.63)	(-11.03)	(-10.60)	(-8.78)	(-8.54)	(-8.23)	(-7.91)	(-6.99)	(-6.71)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	610,212	610,212	610,212	610,212	610,212	610,212	610,212	610,212	610,212	610,212	610,212	610,212
R^2	0.007	0.006	0.007	0.006	0.011	0.010	0.011	0.011	0.010	0.009	0.011	0.010
Test of Marginal Effect:												
Insider Info Subsequent to Upgrade			1.400***				3.073***				3.443***	
			(6.96)				(8.99)				(6.78)	
Insider Info Subsequent to Downgrade			2.066***				3.031***				4.300***	
			(9.79)				(8.75)				(8.34)	
Insider Info Subsequent to No Rec Change			1.542***				3.245***				4.089***	
			(12.75)				(14.63)				(11.86)	
Analyst Info Subsequent to Insider Buy				-0.019				-0.185				0.175
				(-0.08)				(-0.47)				(0.31)
Analyst Info Subsequent to Insider Sell				0.259**				0.318				0.167
				(2.11)				(1.56)				(0.60)
Analyst Info Subsequent to No Insider Trade				0.157**				0.159				0.366**
				(2.20)				(1.29)				(2.11)

Table 9
Regressions of cumulative monthly characteristic-adjusted stock returns (in %): Before and after EA months. The table summarizes additional return regression results using the two subsamples that include observations before and after EA months, respectively. The dependent variable is cumulative monthly characteristic-adjusted abnormal returns over three subsequent holding periods of one, three, and six months (1M, +3M, and +6M) following the two months in which we measure recommendation changes and then insider trading activity, or vice versa. All explanatory variables are lagged variables. Insider Buy (Insider Sell) is defined as the ratio of the value of insider purchases (insider sales) to the value of all insider trades. Net Insider Trade is Insider Buy minus Insider Sell. The analyst recommendations are re-coded as follows: a rating of '1' reflects a strong sell recommendation, '2' a sell, '3' a hold, '4' a buy, and '5' a strong buy recommendation. Rec Change is consensus analyst recommendation change either from month T-2 to T-1 or from T-1 to T, depending on whether analyst recommendation change or insider trading is measured first. Upgrade (Downgrade) is an indicator variable if Recommendation Change is positive (negative). Previous month return is the characteristics-adjusted abnormal return in month T-1. MOM is the firm's past twelve-month characteristics-adjusted abnormal return. Market value of equity (MV) is price per share times total shares outstanding and the book-to-market ratio is measured at the end of previous fiscal year. Volatility is the average daily stock return volatility in the past month. Turnover is the average daily trading volume divided by total shares outstanding in the past month. Institutional holdings is normalized by the total number of shares outstanding. Analyst Coverage is a ranking variable (1 to 5) based on the number of analysts covering the stock in the previous year. Institutional holdings are institutional ownership measured in the previous quarter. % Top 10 Brokers is the percentage of analyst recommendations from the top ten brokers for each firm in a month. Control variables are included in the estimates, but their coefficients are not reported in the table. Industry fixed effects (FE) are based on Fama-French 30-industry groupings, and year FEs are yearly indicator variables during the sample period except for year 2016 to ensure correct model specifications. $T-$statistics, reported in parentheses, are associated with standard errors adjusted for firm-level clustering. (***), (**), and (*) indicate significance at the 1%, 5%, and 10% levels, respectively.

Panel A. +1M								
	Panel A.1. Before EA Month				Panel A.2. After EA Month			
	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
Insider Buy	1.763***				1.077***			
	(10.79)				(10.05)			
Insider Sell	-0.487***				-0.205***			
	(-5.77)				(-3.21)			
Upgrade		-0.010				0.012		
		(-0.12)				(0.17)		
Downgrade		-0.098				-0.309***		
		(-1.18)				(-4.24)		
Analyst Recommendations are Measured First:								
$(\text{Upgrade})_{T-1}$*$(\text{Insider Buy})_T$			1.478***				0.854***	
			(4.71)				(3.58)	
$(\text{Upgrade})_{T-1}$*$(\text{Insider Sell})_T$			-0.428***				-0.190*	
			(-2.69)				(-1.67)	
$(\text{Downgrade})_{T-1}$*$(\text{Insider Buy})_T$			2.108***				1.121***	
			(6.17)				(4.63)	
$(\text{Downgrade})_{T-1}$*$(\text{Insider Sell})_T$			-0.515***				-0.406***	
			(-3.16)				(-3.77)	
$(\text{No Rec Change})_{T-1}$*$(\text{Insider Buy})_T$			1.753***				1.124***	
			(8.07)				(8.66)	
$(\text{No Rec Change})_{T-1}$*$(\text{Insider Sell})_T$			-0.505***				-0.127	
			(-4.63)				(-1.57)	
Insider Trades are Measured First:								
$(\text{Insider Buy})_{T-1}$*$(\text{Upgrade})_T$				1.262***				0.270
				(4.69)				(1.13)
$(\text{Insider Buy})_{T-1}$*$(\text{Downgrade})_T$				0.870***				-0.035
				(3.15)				(-0.15)
$(\text{Insider Sell})_{T-1}$*$(\text{Upgrade})_T$				-0.423***				-0.202
				(-3.01)				(-1.60)
$(\text{Insider Sell})_{T-1}$*$(\text{Downgrade})_T$				-0.675***				-0.560***
				(-4.82)				(-4.42)
$(\text{No Trade})_{T-1}$*$(\text{Upgrade})_T$				-0.034				0.033
				(-0.37)				(0.42)
$(\text{No Trade})_{T-1}$*$(\text{Downgrade})_T$				-0.029				-0.284***
				(-0.31)				(-3.50)
Intercept	-1.183***	-1.093***	-1.149***	-1.120***	-2.529***	-2.425***	-2.580***	-2.436***
	(-3.48)	(-3.21)	(-3.36)	(-3.29)	(-8.80)	(-8.45)	(-8.97)	(-8.48)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	263,717	263,717	263,717	263,717	263,846	263,846	263,846	263,846
R^2	0.013	0.012	0.013	0.012	0.007	0.006	0.007	0.006
Test of Marginal Effect:								
Insider Info Subsequent to Upgrade			1.907***				1.044***	
			(5.41)				(4.03)	
Insider Info Subsequent to Downgrade			2.623***				1.527***	
			(7.03)				(5.92)	
Insider Info Subsequent to No Rec Change			2.259***				1.251***	
			(9.47)				(8.54)	
Analyst Info Subsequent to Insider Buy				0.391				0.305
				(1.02)				(0.93)
Analyst Info Subsequent to Insider Sell				0.252				0.359**
				(1.37)				(2.17)
Analyst Info Subsequent to No Insider Trade				-0.005				0.317***
				(-0.04)				(3.43)

(*continued on next page*)

Table 9 (*continued*)

Panel B. +3M	Panel B.1. Before EA Month				Panel B.2. After EA Month			
	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
Insider Buy	2.166***				2.517***			
	(9.05)				(13.41)			
Insider Sell	-0.705***				-0.783***			
	(-5.30)				(-7.15)			
Upgrade		-0.327**				-0.066		
		(-2.55)				(-0.53)		
Downgrade		-0.259**				-0.441***		
		(-1.96)				(-3.46)		
Analyst Recommendations are Measured First:								
$(\text{Upgrade})_{T-1}$*$(\text{Insider Buy})_T$			2.405***				2.165***	
			(4.44)				(5.46)	
$(\text{Upgrade})_{T-1}$*$(\text{Insider Sell})_T$			-0.780***				-0.788***	
			(-3.18)				(-4.07)	
$(\text{Downgrade})_{T-1}$*$(\text{Insider Buy})_T$			1.734***				2.059***	
			(3.52)				(5.16)	
$(\text{Downgrade})_{T-1}$*$(\text{Insider Sell})_T$			-0.726***				-1.094***	
			(-2.93)				(-5.66)	
$(\text{No Rec Change})_{T-1}$*$(\text{Insider Buy})_T$			2.244***				2.746***	
			(7.30)				(11.70)	
$(\text{No Rec Change})_{T-1}$*$(\text{Insider Sell})_T$			-0.693***				-0.663***	
			(-3.90)				(-4.73)	
Insider Trades are Measured First:								
$(\text{Insider Buy})_{T-1}$*$(\text{Upgrade})_T$				1.560***				1.098***
				(3.76)				(2.66)
$(\text{Insider Buy})_{T-1}$*$(\text{Downgrade})_T$				1.946***				0.807*
				(4.49)				(1.94)
$(\text{Insider Sell})_{T-1}$*$(\text{Upgrade})_T$				-0.881***				-0.721***
				(-4.15)				(-3.25)
$(\text{Insider Sell})_{T-1}$*$(\text{Downgrade})_T$				-1.307***				-1.051***
				(-6.07)				(-4.64)
$(\text{No Trade})_{T-1}$*$(\text{Upgrade})_T$				-0.382***				-0.038
				(-2.62)				(-0.28)
$(\text{No Trade})_{T-1}$*$(\text{Downgrade})_T$				-0.178				-0.437***
				(-1.16)				(-3.08)
Intercept	-4.323***	-4.287***	-4.264***	-4.327***	-4.700***	-4.496***	-4.917***	-4.524***
	(-7.57)	(-7.47)	(-7.48)	(-7.55)	(-8.14)	(-7.68)	(-8.48)	(-7.72)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	263,717	263,717	263,717	263,717	263,846	263,846	263,846	263,846
R^2	0.018	0.017	0.018	0.017	0.009	0.008	0.009	0.009
Test of Marginal Effect:								
Insider Info Subsequent to Upgrade			3.185***				2.953***	
			(5.37)				(6.76)	
Insider Info Subsequent to Downgrade			2.460***				3.153***	
			(4.53)				(7.12)	
Insider Info Subsequent to No Rec Change			2.937***				3.409***	
			(8.46)				(12.98)	
Analyst Info Subsequent to Insider Buy				0.391				0.291
				(1.02)				(0.50)
Analyst Info Subsequent to Insider Sell				0.252				0.330
				(1.37)				(1.13)
Analyst Info Subsequent to No Insider Trade				-0.005				0.400**
				(-0.04)				(2.45)

Panel C. +6M	Panel C.1. Before EA Month				Panel C.2. After EA Month			
	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
Insider Buy	2.761***				3.434***			
	(7.86)				(11.86)			
Insider Sell	-0.936***				-0.519***			
	(-4.57)				(-3.06)			
Upgrade		-0.292				-0.003		
		(-1.59)				(-0.02)		
Downgrade		-0.432**				-0.431**		
		(-2.29)				(-2.37)		

(*continued on next page*)

Table 9 (*continued*)

	Panel C.1. Before EA Month				Panel C.2. After EA Month			
	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
Analyst Recommendations are Measured First:								
$(\text{Upgrade})_{T-1}$*$(\text{Insider Buy})_T$			2.181***				2.738***	
			(2.84)				(4.79)	
$(\text{Upgrade})_{T-1}$*$(\text{Insider Sell})_T$			-1.290***				-0.420	
			(-3.69)				(-1.48)	
$(\text{Downgrade})_{T-1}$*$(\text{Insider Buy})_T$			2.133***				3.491***	
			(3.04)				(5.91)	
$(\text{Downgrade})_{T-1}$*$(\text{Insider Sell})_T$			-0.629*				-1.246***	
			(-1.71)				(-4.44)	
$(\text{No Rec Change})_{T-1}$*$(\text{Insider Buy})_T$			3.109***				3.604***	
			(7.23)				(10.00)	
$(\text{No Rec Change})_{T-1}$*$(\text{Insider Sell})_T$			-0.942***				-0.284	
			(-3.57)				(-1.34)	
Insider Trades are Measured First:								
$(\text{Insider Buy})_{T-1}$*$(\text{Upgrade})_T$				2.635***				1.824***
				(4.40)				(3.14)
$(\text{Insider Buy})_{T-1}$*$(\text{Downgrade})_T$				2.292***				1.163*
				(3.80)				(1.93)
$(\text{Insider Sell})_{T-1}$*$(\text{Upgrade})_T$				-0.930***				-0.619*
				(-3.06)				(-1.94)
$(\text{Insider Sell})_{T-1}$*$(\text{Downgrade})_T$				-1.272***				-0.416
				(-4.12)				(-1.31)
$(\text{No Trade})_{T-1}$*$(\text{Upgrade})_T$				-0.433**				-0.042
				(-2.05)				(-0.21)
$(\text{No Trade})_{T-1}$*$(\text{Downgrade})_T$				-0.472**				-0.596***
				(-2.15)				(-2.93)
Intercept	-5.512***	-5.462***	-5.472***	-5.494***	-6.318***	-5.913***	-6.572***	-5.921***
	(-5.43)	(-5.36)	(-5.38)	(-5.39)	(-5.92)	(-5.54)	(-6.29)	(-5.54)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	263,717	263,717	263,717	263,717	263,846	263,846	263,846	263,846
R^2	0.014	0.014	0.015	0.014	0.008	0.007	0.008	0.008
Test of Marginal Effect:								
Insider Info Subsequent to Upgrade			3.470***				3.158***	
			(4.12)				(4.95)	
Insider Info Subsequent to Downgrade			2.763***				4.738***	
			(3.52)				(7.33)	
Insider Info Subsequent to No Rec Change			4.051***				3.888***	
			(8.24)				(9.75)	
Analyst Info Subsequent to Insider Buy				0.344				0.662
				(0.41)				(0.81)
Analyst Info Subsequent to Insider Sell				0.342				-0.203
				(0.89)				(-0.50)
Analyst Info Subsequent to No Insider Trade				0.039				0.554**
				(0.15)				(2.38)

before the EA month. Second, when evaluated conditionally, insider trading remains informative following analyst recommendation revisions in both periods. Analyst downgrades are also informative following insider selling, but such impact only occur after the EA month. Third, the joint impact of two concurring signals (insider selling and analyst downgrades) continues to be more severe than that of either of the signals alone, whereas the impact of two contrasting signals (insider buying and analyst downgrades) tends to fall between that of negative and positive signals alone. Fourth, the singular substitution effect of insider buying following analyst downgrades is stronger before than after the EA month: 2.11% (model 3) vs. 1.12% (model 7). Overall, the combined findings not only support the complementarity effect of insider trading and analyst downgrades as well as the substitution effect of insider buying following analyst downgrades, but also imply that the return predictability of insider trading and analyst downgrades stems from private and public information, respectively.

7. Summary

This paper first examines the dynamic interactions between insiders and analysts. The results reveal that insiders are likely to purchase more shares when their firm's stock is downgraded by analysts but sell more when upgraded. The contrasting insider trading pattern, vis-*â*-vis analyst recommendation revisions, persists even after controlling for insider and analyst stock preferences, various analyst and stock characteristics, as well as the overall information environment. Thus, it is unlikely that our evidence is merely driven by insiders exploiting opportunistic mispricing by buying after stock price declines or selling after stock price increases. Rather, further analysis shows that insiders' motives are likely driven by signaling rather than information reasons as insider buying significantly increases following analyst downgrades, particularly those associated with more severe return consequences. On the other hand, although there is some association between insider buying and subsequent analyst upgrades, we do not find a clear pattern that analysts systematically incorporate insider trading information when revising their recommendations.

Next, we investigate the informativeness of insider trading and analyst recommendations and more importantly, test the complementarity vs. substitution hypotheses. When evaluated separately, both insider trading and analyst recommendations are informative for future stock returns. For insiders, we find that both their buying and selling contain information, but the former is economically more significant than the latter. Top insiders' trades also contain more value-relevant information than other insiders. For analysts, we find that only their downgrades, but not upgrades, are associated with considerable investment value. As such, we focus mainly on the relative informativeness of analyst downgrades and insider trading (buying and selling) in subsequent analyses. The results show that the return predictability of insider trading conditional on prior analyst downgrades remains significant, and analyst downgrades remain informative conditional on prior insider selling. Further, the joint informativeness of these two signals is more severe when they concur (i.e., insider selling and analyst downgrades) but attenuated when they disagree (insider buying and analyst downgrades), implying the effectiveness (and complementarity) of these two signals. Thus, the combined evidence lends support for the complementarity hypothesis. However, our results also reveal the strong signaling effect of insider buying following analyst downgrades. Extended analyses confirm both effects and further suggest that analysts provide value-pertinent information to the markets through analyzing public information while insiders are likely to exploit their private information. Collectively, the return results indicate that both insider trading and analyst recommendations could provide complementary information to the markets although insider buying following analyst downgrades could be singularly dominant and substitute the negative information effect of those downgrades. A clear implication from our combined findings is that although both insider trading and analyst recommendations are valuable signals, it is important for investors to discern the sequences and relative informativeness of these signals while also recognize the strong signaling of insider buying under certain circumstances.

CRediT authorship contribution statement

Jim Hsieh: Conceptualization, Methodology, Formal analysis, Writing – original draft, Project administration. **Lilian Ng:** Conceptualization, Methodology, Formal analysis, Writing – original draft. **Qinghai Wang:** Conceptualization, Methodology, Formal analysis, Writing – original draft.

Data availability

The authors do not have permission to share data.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jbankfin.2023.106787.

References

Agrawal, A., Chadha, S., Chen, M.A., 2006. Who is afraid of reg FD? the behavior and performance of sell-side analysts following the SEC's fair disclosure rules. Journal of Business 79, 2811–2834.
Alldredge, D.M., Cicero, D.C., 2015. Attentive insider trading. J Financ Econ 115, 84–101.
Altinkilic, O., Hansen, R.S., 2009. On the information role of stock recommendation revisions. Journal of Accounting and Economics 48, 17–36.
Barber, B.M., Lehavy, R., McNichols, M., Trueman, B., 2001. Can investors profit from the prophets? security analyst recommendations and stock returns. Journal of Finance 56, 531–563.
Bettis, J.C., Coles, J.L., Lemmon, M.L., 2000. Corporate policies restricting trading by insiders. J Financ Econ 57, 191–220.
Bidwell III, C.M., 1977. How good is institutional brokerage research? Journal of Portfolio Management 3, 26–31.
Boehmer, E., Jones, C.M., Wu, J.J., Zhang, X., 2020. What do short sellers know? Rev Financ 24, 1203–1235.
Bradley, D., Clarke, J., Lee, S., Ornthanalai, C., 2014. Are analysts' recommendations informative? intraday evidence on the impact of time stamp delays. Journal of Finance 69, 645–673.
Bradshaw, M.T., 2004. How do analysts use their earnings forecasts in generating stock recommendations? The Accounting Review 79, 25–50.
Brochet, F., 2010. Information content of insider trades before and after the sarbanes-oxley act. The Accounting Review 85, 419–446.
Carpenter, J.N., Remmers, B., 2001. Executive option exercises and inside information. Journal of Business 74, 513–534.
Cheng, Q., Lo, K., 2006. Insider trading and voluntary disclosures. Journal of Accounting Research 44, 815–848.
Christophe, S.E., Ferri, M.G., Hsieh, J., 2010. Informed trading before analyst downgrades: evidence from short sellers. J Financ Econ 95, 85–106.
Christophe, S.E., Hsieh, J., 2023. Information percolation and informed short selling: evidence from earnings announcements. Working Paper.
Cowen, A., Groysberg, B., Healy, P., 2006. Which types of analyst firms are more optimistic? Journal of Accounting and Economics 41, 119–146.
Damodaran, A., Liu, C.H., 1993. Insider trading as a signal of private information. Review of Financial Studies 6, 79–119.
Daniel, K.D., Titman, S., 1997. Evidence on the characteristics of cross-sectional variation in stock returns. Journal of Finance 52, 1–33.
Das, S., Levine, C.B., Sivaramakrishnan, K., 1998. Earnings predictability and bias in analysts' earnings forecasts. Accounting Review 73, 277–294.
Dimson, E., Marsh, P., 1984. An analysis of brokers' and analysts' unpublished forecasts of UK stock returns. Journal of Finance 39, 1257–1292.
Ellul, A., Panayides, M., 2018. Do financial analysts restrain insiders' informational advantage? Journal of Financial and Quantitative Analysis 53, 203–241.
Elton, J.E., Gruber, M.J., Grossman, S., 1986. Discrete expectational data and portfolio performance. Journal of Finance 41, 699–713.
Engelberg, J., McLean, R.D., Pontiff, J., 2018. Analysts and anomalies. Working Paper.
Engelberg, J.E., Reed, A.V., Ringgenberg, M.C., 2012. How are shorts informed? short sellers, news, and information processing. J Financ Econ 105, 260–278.
Finnerty, J., 1976. Insiders and market efficiency. Journal of Finance 31, 1141–1148.
Fishman, M.J., Hagerty, K.M., 1992. Insider trading and efficiency of stock prices. Rand J Econ 23, 106–122.
Francis, J., Philbrick, D.R., 1993. Analysts' decisions as products of a multi-task environment. Journal of Accounting Research 31, 216–230.
Francis, J., Venkatachalam, M., Zhang, Y., 2005. Do short sellers convey information about changes in fundamentals or risk?Working Paper, Duke University.
Frankel, R., Li, X., 2004. Characteristics of a firm's information environment and the information asymmetry between insiders and outsiders. Journal of Accounting and Economics 37, 229–259.
George, T., Seyhun, H. N., 2002. Does insider trading enhances market efficiency. Working paper, University of Houston and University of Michigan.
Gintschel, A., Markov, S., 2004. The effectiveness of regulation FD. Journal of Accounting and Economics 37, 293–314.
Groth, J.C., Lewellen, W.G., Schlarbaum, G.G., Lease, R.C., 1979. An analysis of brokerage house securities recommendations. Financial Analysts Journal 35, 32–40.
Guo, L., Li, F.W., Wei, K.C.J., 2019. Security analysts and capital market anomalies. J Financ Econ. Forthcoming
Hartzell, J.C., Starks, L.T., 2003. Institutional investors and executive compensation. Journal of Finance 58, 2351–2374.
Howe, J.S., Unlu, E., Yan, X.S., 2009. The predictive content of aggregate analyst recommendations. Journal of Accounting Research 47, 799–822.
Huddart, S., Ke, B., Shi, C., 2007. Jeopardy, non-public information, and insider trading around SEC 10-K and 10-Q filings. Journal of Accounting and Economics 43, 3–36.
Hutton, A.P., Lee, L.F., Shu, S.Z., 2012. Do managers always know better? the relative accuracy of management and analyst forecasts. Journal of Accounting Research 50, 1217–1244.
Jaffe, J., 1974. Special information and insider trading. Journal of Business 47, 410–428.
Jegadeesh, N., Kim, J., Krische, S.D., Lee, C.M.C., 2004. Analyzing the analysts: when do recommendations add value? Journal of Finance 59, 1083–1124.
Jeng, L., Metrick, A., Zeckhauser, R., 2003. Estimating the returns to insider trading: aperformance-evaluation perspective. Rev Econ Stat 85, 453–471.
Jenter, D., 2005. Market timing and managerial portfolio decisions. Journal of Finance 60, 1903–1949.
Kadan, O., Madureira, L., Wang, R., Zach, T., 2009. Conflicts of interest and stock recommendations: the effects of the global settlement and related regulations. Review of Financial Studies 22, 4189–4217.
Kadan, O., Madureira, L., Wang, R., Zach, T., 2012. Analysts' Industry expertise. Journal of Accounting and Economics 54, 95–120.
Kallunki, J.-P., Nilsson, H., Hellstrom, J., 2009. Why do insiders trade? evidence based on unique data on swedish insiders. Journal of Accounting and Economics 48, 37–53.
Karpoff, J.M., Lou, X., 2010. Short sellers and financial misconduct. Journal of Finance 65, 1879–1913.
Khanna, N., Slezak, S.L., Bradley, M., 1994. Insider trading, outside search, and resource allocation: why firms and society may disagree on insider trading restrictions. Review of Financial Studies 7, 575–608.
Kolasinski, A.C., Kothari, S.P., 2008. Investment banking and analyst objectivity: evidence from analysts affiliated with mergers and acquisitions advisors. Journal of Financial and Quantitative Analysis 43, 817–842.
Korczak, A., Korczak, P., Lasfer, M., 2010. To trade or not to trade, the strategic trading of insiders around news announcements. Journal of Business Finance & Accounting 37, 369–407.
Lakonishok, J., Lee, I., 2001. Are insider trades informative? Review of Financial Studies 14, 79–111.

Leland, H.E., 1992. Insider trading: should it be prohibited? Journal of Political Economy 100, 859–887.
Li, E.X., Ramesh, K., Shen, M., Wu, J.S., 2015. Do analyst stock recommendations piggyback on recent corporate news? an analysis of regular-hour and after-hours revisions. Journal of Accounting Research 53, 821–861.
Lin, H.-w., McNichols, M., 1998. Underwriting relationships, analysts' earnings forecasts and investment recommendations. Journal of Accounting and Economics 25, 101–127.
Ljungqvist, A., Marston, F., Starks, L.T., Wei, K.D., Yan, H., 2007. Conflicts of interest in sell-side research and the moderating role of institutional investors. J Financ Econ 85, 420–456.
Loh, R.K., Stulz, R.M., 2011. When are analyst recommendation changes influential? Review of Financial Studies 24, 593–627.
Loh, R.K., Stulz, R.M., 2018. Is sell-side research more valuable in bad times? Journal of Finance 73, 959–1013.
Loureiro, G., Taboada, A.G., 2015. Do improvements in the information environment enhance insiders' ability to learn from outsiders? Journal of Accounting Research 53, 863–905.
Lustgarten, S., Mande, V., 1995. Financial analysts' earnings forecasts and insider trading. Journal of Accounting and Public Policy 14, 233–261.
Lustgarten, S., Mande, V., 1998. The effect of insider trading on financial analysts' forecast accuracy and dispersion. Journal of Accounting and Public Policy 17, 311–327.
Marin, J.M., Olivier, J.P., 2008. The dog that did not bark: insider trading and crashes. Journal of Finance 63, 2429–2476.
McVay, S., Nagar, V., Tang, V.W., 2006. Trading incentives to meet the analyst forecast. Review of Accounting Studies 11, 575–598.
Michaely, R., Womack, K., 1999. Conflict of interest and the credibility of underwriter analyst recommendations. Review of Financial Studies 653–686.
Ofek, E., Yermack, D., 2000. Taking stock: equity-based compensation and evolution of managerial ownership. Journal of Finance 55, 1367–1384.
Piotroski, J.D., Roulstone, D., 2004. The influence of analysts, institutional investors and insiders on the incorporation of market, industry and firm-specific information into stock prices. The Accounting Review 79, 1119–1151.
Rozeff, M.S., Zaman, M.A., 1998. Overreaction and insider trading: evidence from growth and value portfolios. Journal of Finance 53, 701–716.
Sawicki, J., Shrestha, K., 2008. Insider trading and earnings management. Journal of Business Finance & Accounting 35, 331–346.
Seyhun, N., 1986. Insiders' profits, costs of trading, and market efficiency. J Financ Econ 47, 410–428.
Shleifer, A., Vishny, R.W., 1986. Large shareholders and corporate control. Journal of Political Economy 94, 461–488.
Sivakumar, K., Vijayakumar, J., 2001. Insider trading, analysts' forecast revisions, and earnings changes. Journal of Accounting, Auditing, and Finance 16, 167–187.
Stickel, S.E., 1995. The anatomy of the performance of buy and sell recommendations. Financial Analysts Journal 25–39.
Wang, S., 2019. Informational environments and the relative information content of analyst recommendations and insider trades. Accounting, Organizations and Society 72, 61–73.
Womack, K., 1996. Do brokerage analysts' recommendations have investment value? Journal of Finance 51, 137–167.
Wu, W., 2016. Information asymmetry and insider trading. Working Paper, Texas A&M University.
Yezegel, A., 2015. Why do analysts revise their stock recommendations after earnings announcements? Journal of Accounting and Economics 59, 163–181.